

Lecture 15:
Language Generation & Evaluation

Instructor: Xiang Ren

USC CSCI 544 Applied NLP

Oct 27, Fall 2026

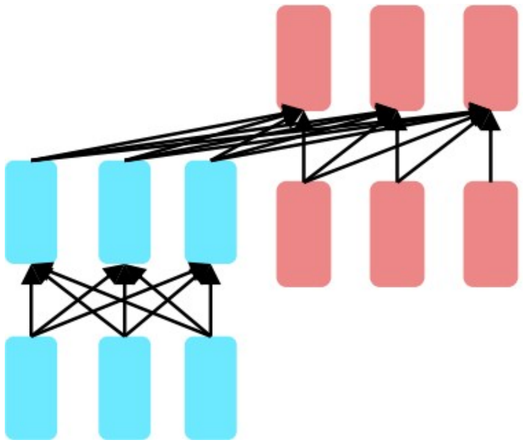
Guest Lecture: Brihi Joshi

Some slides adapted from Dan Jurafsky and Chris Manning

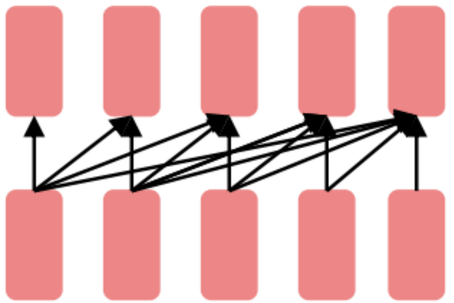
Recap: Broad Spectrum of NLG Tasks

Less Open-Ended — **More Open-Ended**

- Machine Translation
- Summarization
- Task-driven Dialog
- Chitchat Dialog
- Story Generation



Encoder-Decoders



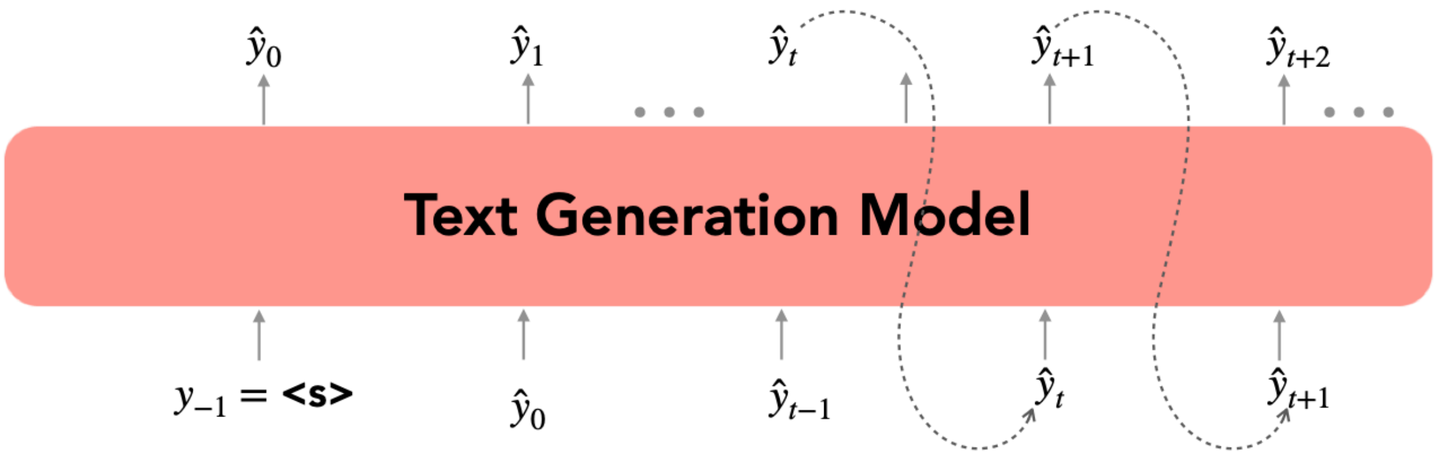
Decoders

Language Generation: Fundamentals

In autoregressive text generation models, at each time step t, our model takes in a sequence of tokens as input $S = f_\theta(y_{<t}) \in \mathbb{R}^V$ and outputs a new token, $\hat{y}_t$

For model $f_\theta(\cdot)$ and vocabulary V, we get scores $S = f_\theta(y_{<t}) \in \mathbb{R}^V$

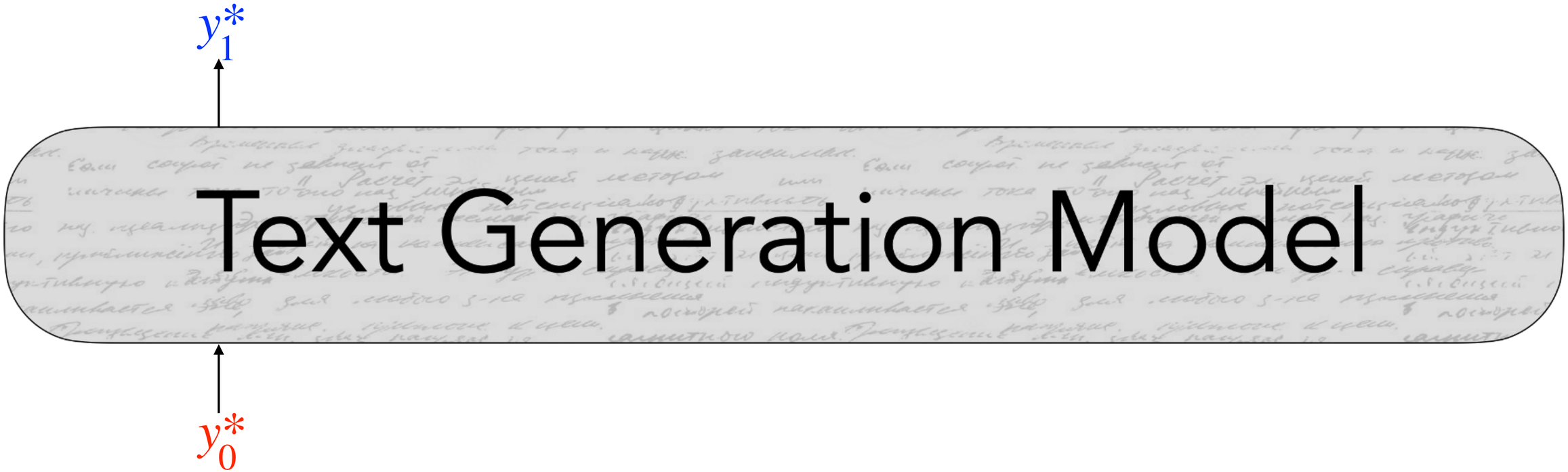
$$P(w \mid y_{<t}) = \frac{\exp(S_w)}{\sum_{v \in V} \exp(S_v)}$$



Maximum Likelihood Training

- Trained to generate the next word y_t^* given a set of preceding words $\{y^*\}_{<t}$

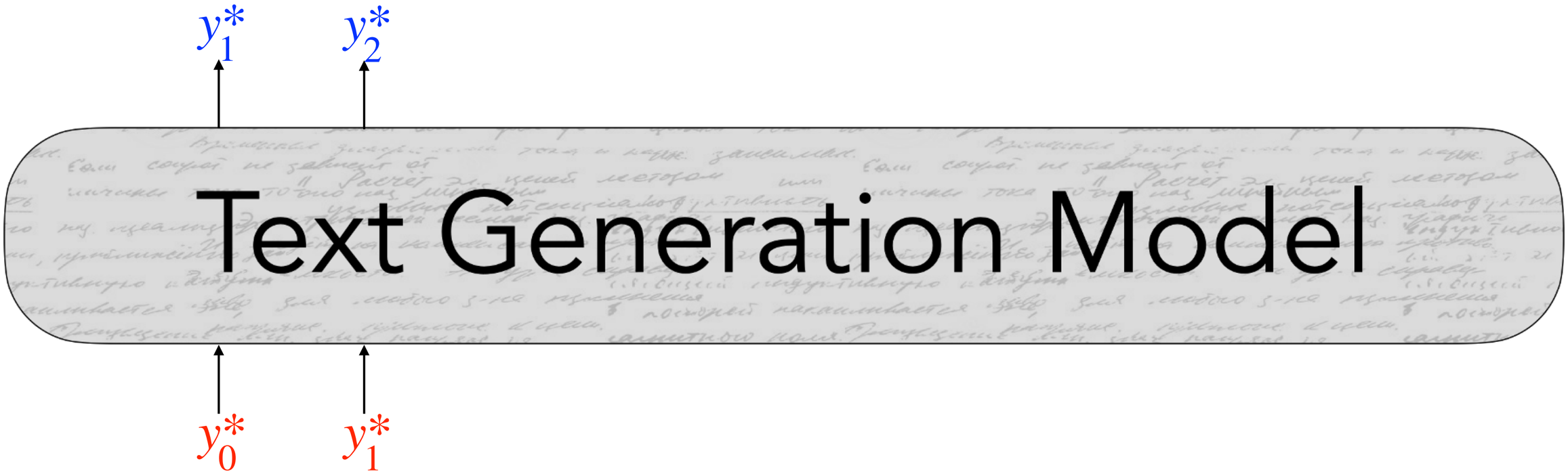
$$L = -\log P(y_1^* \mid y_0^*)$$



Maximum Likelihood Training

- Trained to generate the next word y_t^* given a set of preceding words $\{y^*\}_{<t}$

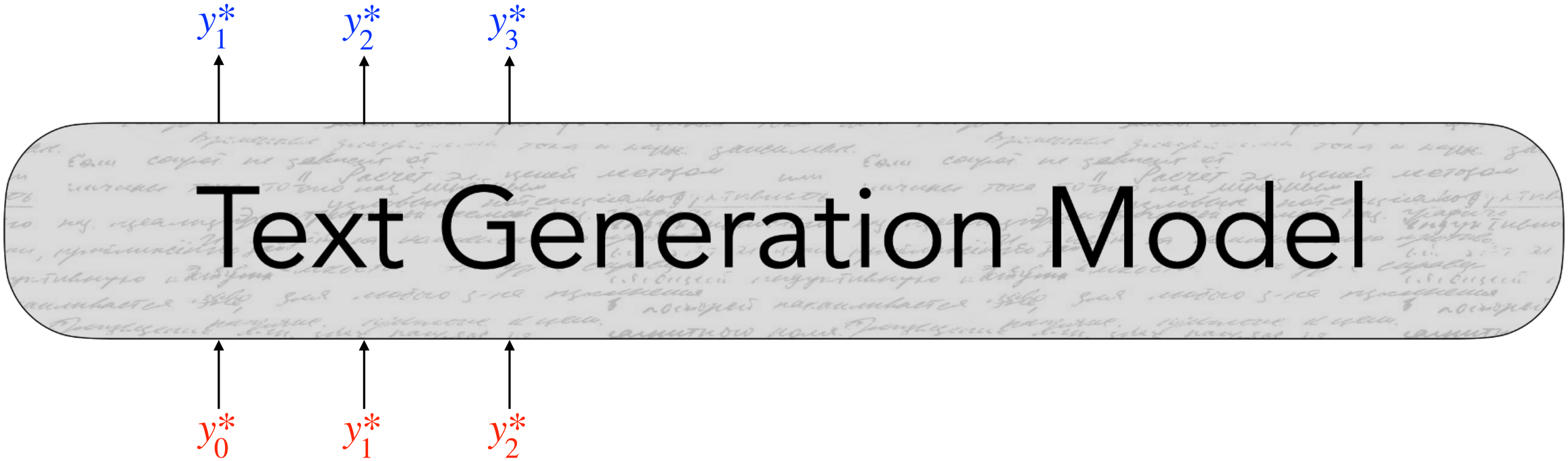
$$L = -\left(\log P(y_1^* \mid y_0^*) + \log P(y_2^* \mid y_0^*, y_1^*)\right)$$



Maximum Likelihood Training

- Trained to generate the next word y_t^* given a set of preceding words $\{y^*\}_{<t}$

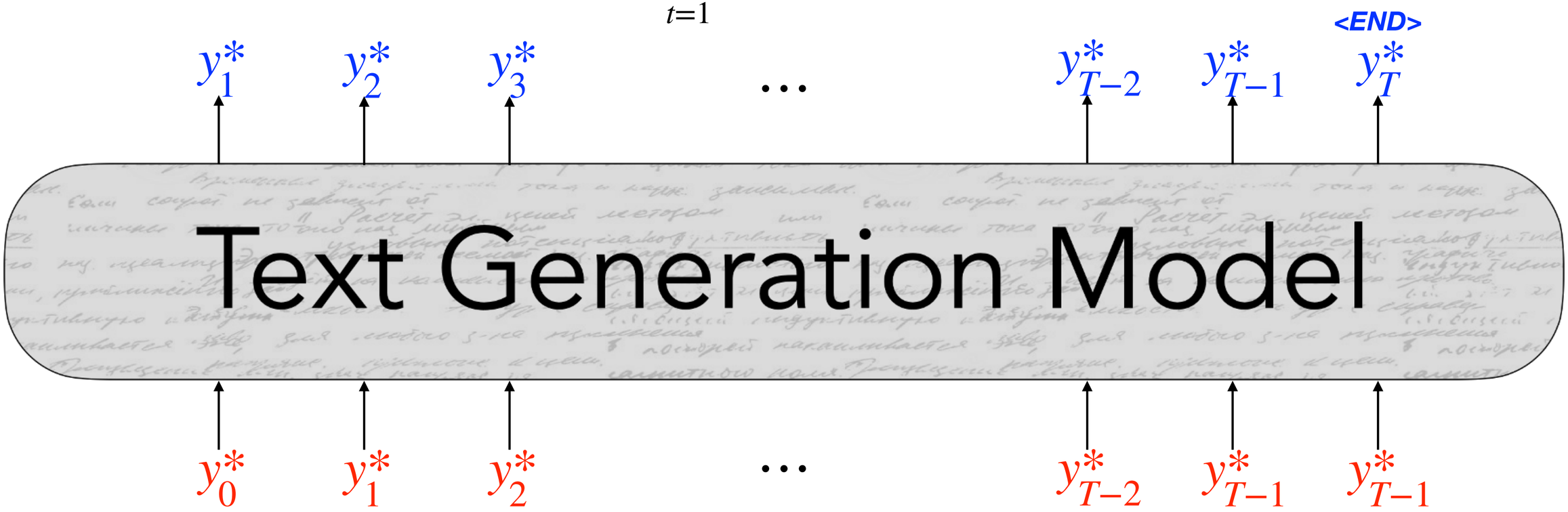
$$L = -\left(\log P(y_1^* \mid y_0^*) + \log P(y_2^* \mid y_0^*, y_1^*) + \log P(y_3^* \mid y_0^*, y_1^*, y_2^*)\right)$$



Maximum Likelihood Training

- Trained to generate the next word y_t^* given a set of preceding words $\{y^*\}_{<t}$

$$L = -\sum_{t=1}^{T} \log P(y_t^* \mid \{y^*\}_{<t})$$

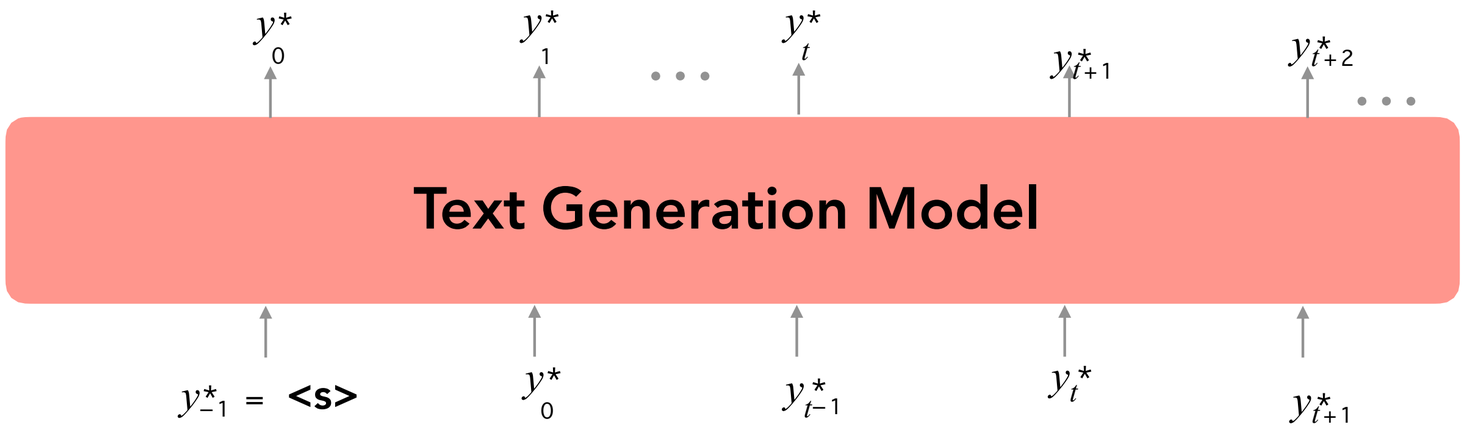


Language Generation: Training

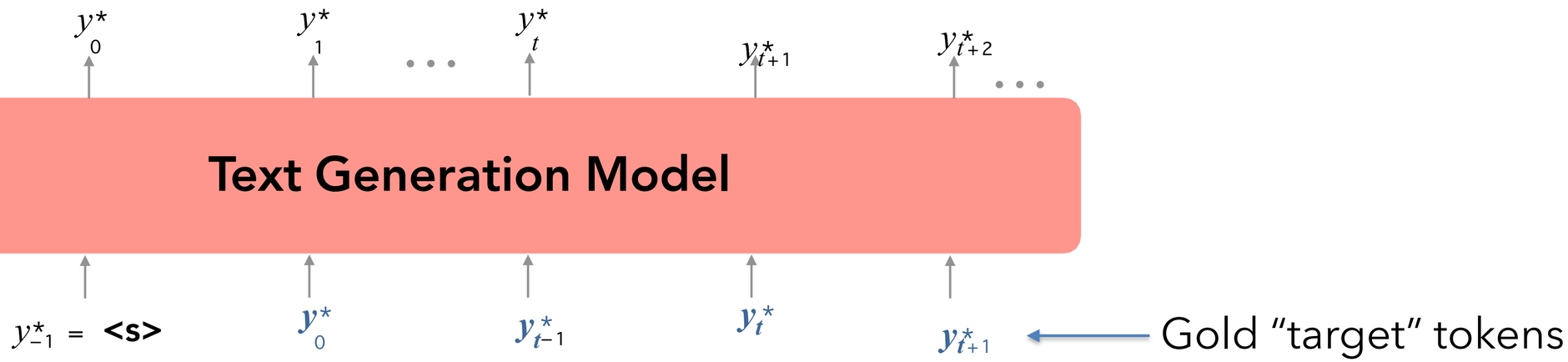
- Trained one token at a time to maximize the probability of the next token y_t^* given preceding words $y_{<t}^*$

$$\mathscr{L} = -\sum_{t=1}^{T} \log P(y_t | y_{<t}) = -\sum_{t=1}^{T} \log \frac{\exp(S_{y_t|y_{<t}})}{\sum_{v \in V} \exp(S_{v|y_{<t}})}$$

- Classification task at each time step trying to predict the actual word y_t^* in the training data
- "Teacher forcing" (reset at each time step to the ground truth)



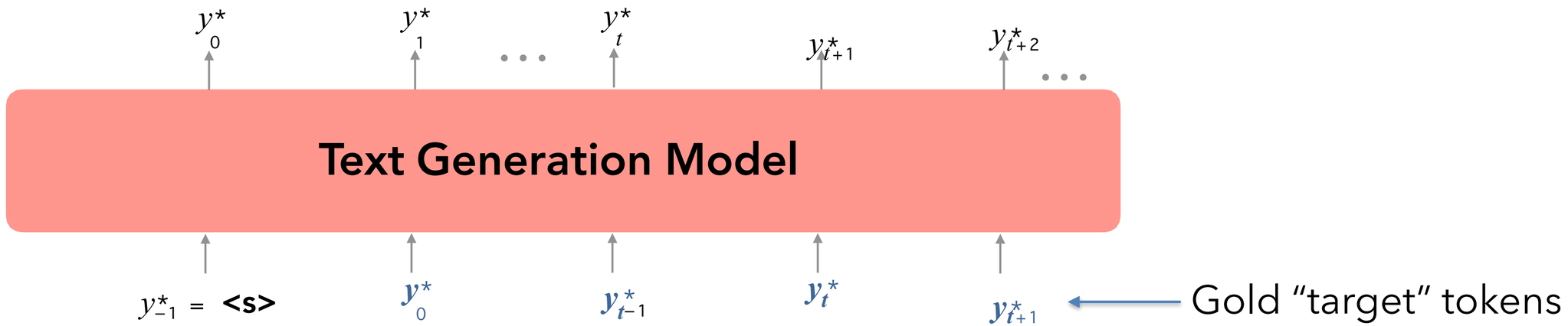
Teacher Forcing



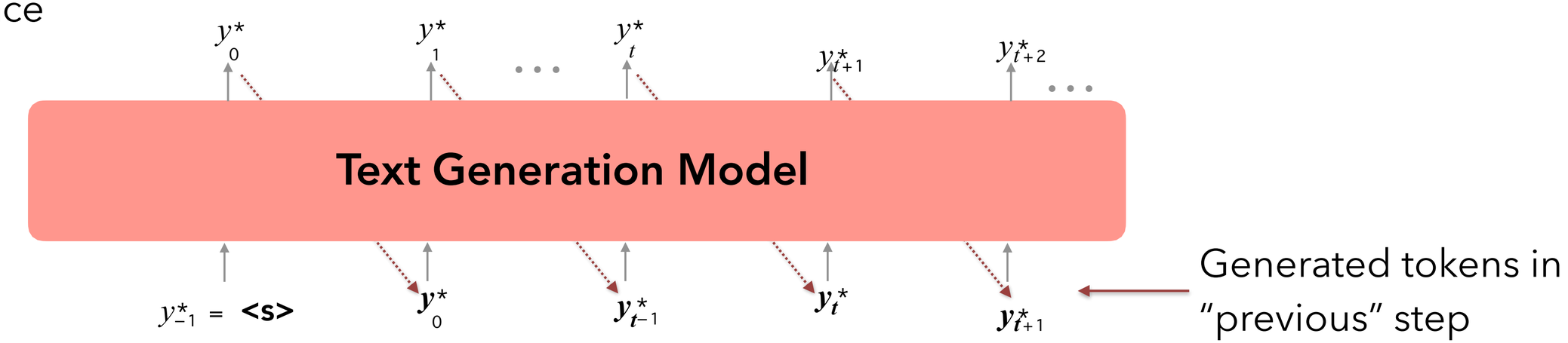
- Strategy for **training** decoders / language models
- At each time step t in decoding we force the system to use the gold target token from training as the next input x_{t+1}, rather than allowing it to rely on the (possibly erroneous) decoder output $\hat{y}_t$
- Runs the risk of **exposure bias**!
  - During training, our model's inputs are gold context tokens from real, human-generated texts
  - At generation time, our model's inputs are previously–decoded tokens

Language Generation: Inference

Training



Inference



Language Generation: Inference

- At each time step *t*, our model computes a vector of scores for each token in our vocabulary, $S \in \mathbb{R}^V$:

$$S = \underbrace{f(\{y_{<t}\}; \theta)}_{f(\cdot\,; \theta) \text{ is your model}}$$

- Then, we compute a probability distribution P over $w \in V$ using these scores:

$$P(y_t = w \mid \{y_{<t}\}) = \frac{\exp(S_w)}{\sum_{w' \in V} \exp(S_{w'})}$$

- Our **decoding** algorithm defines a function to select a token from this distribution:

$$\hat{y}_t = \underbrace{g(P(y_t \mid \{y_{<t}\}))}_{g(\cdot) \text{ is your decoding algorithm}}$$

Note: *we decode token by token from LLMs after they are trained (during inference)*

How to find the most likely text to generate?

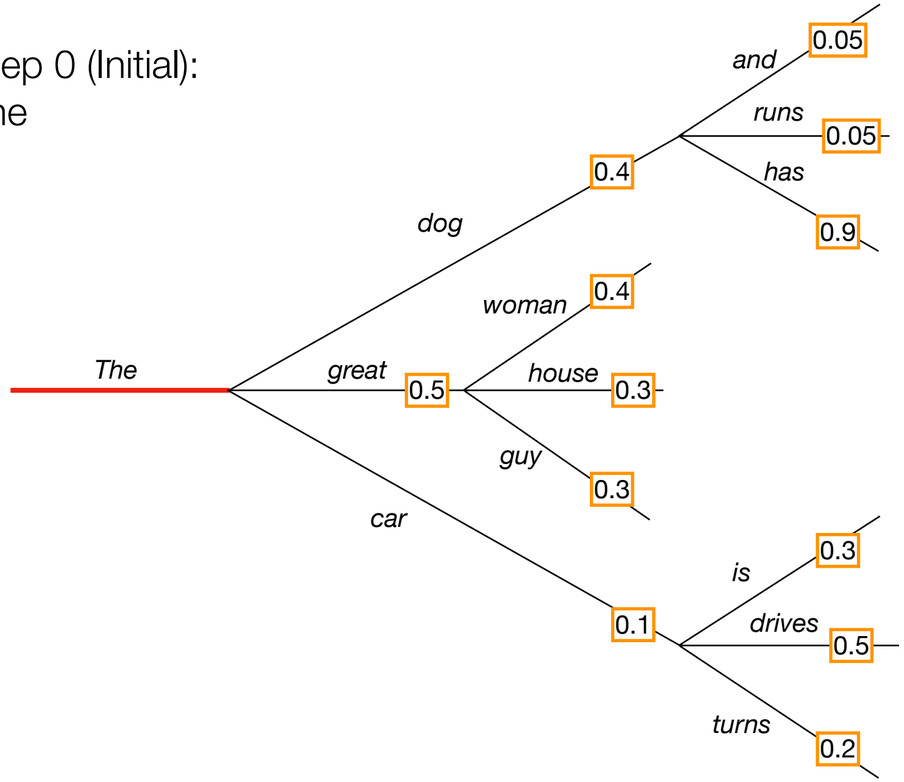
- **Obvious method: Greedy Decoding**
  - Selects the highest probability token according to $P(y_t \mid y_{<t})$

$$\hat{y}_t = \textbf{argmax}_{w \in V} P(y_t = w \mid y_{<t})$$

- **Beam Search**
  - Also aims to find the string with the highest probability, but with a wider exploration of candidates.

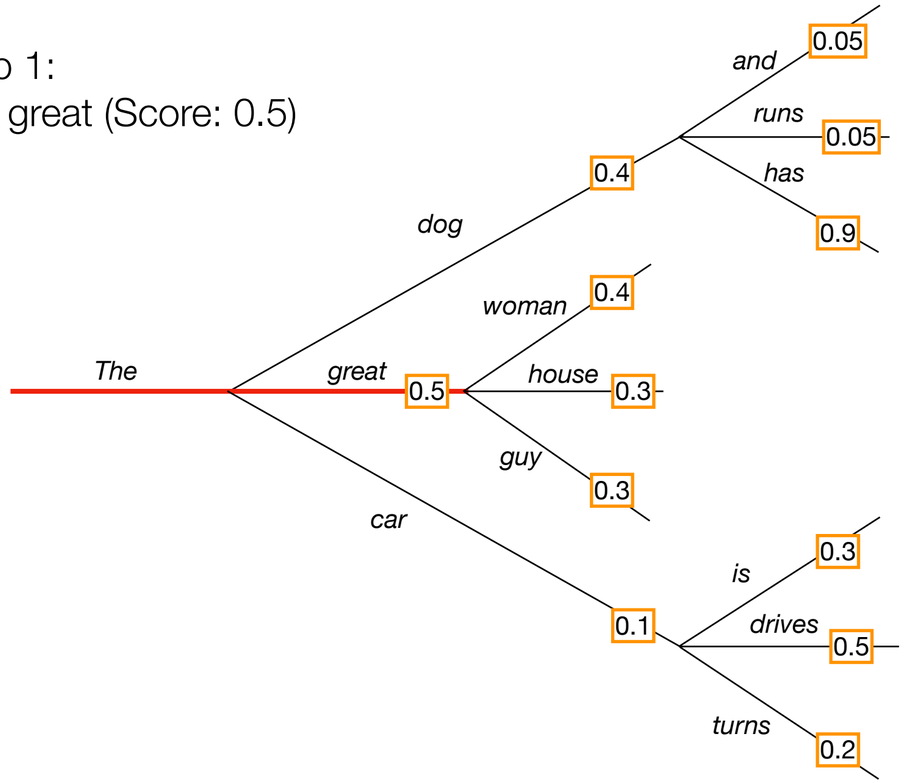
Greedy Decoding vs. Beam Search

- **Greedy Decoding**
  - Choose the "currently best" token at each time step



Greedy Decoding vs. Beam Search

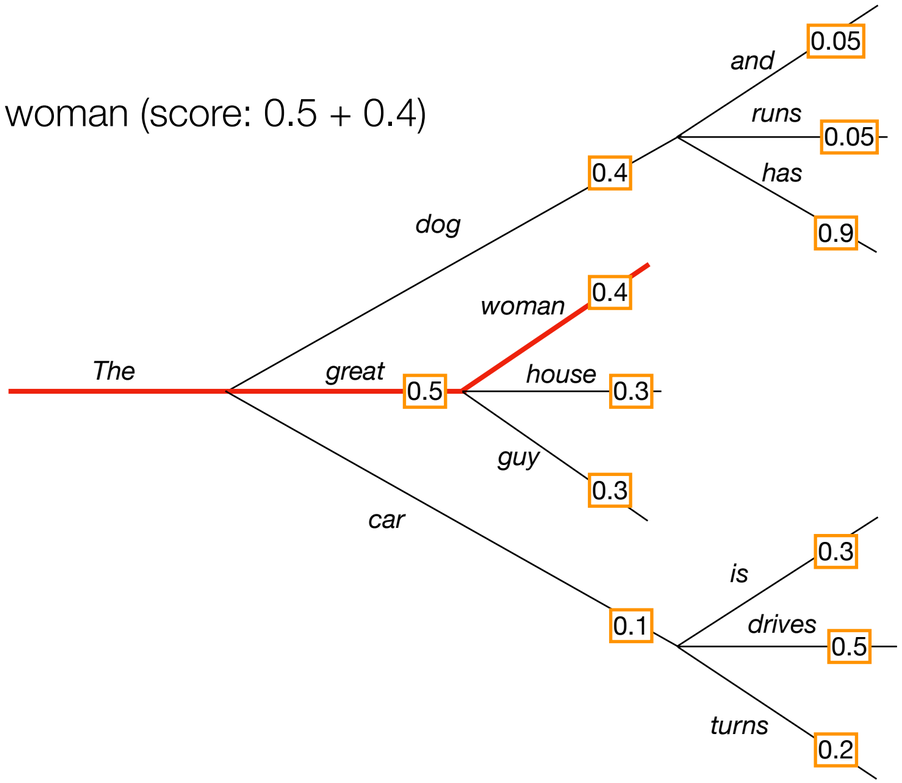
- **Greedy Decoding**
  - Choose the "currently best" token at each time step



Greedy Decoding vs. Beam Search

- **Greedy Decoding**
  - Choose the "currently best" token at each time step

Step 2:
The great woman (score: 0.5 + 0.4)

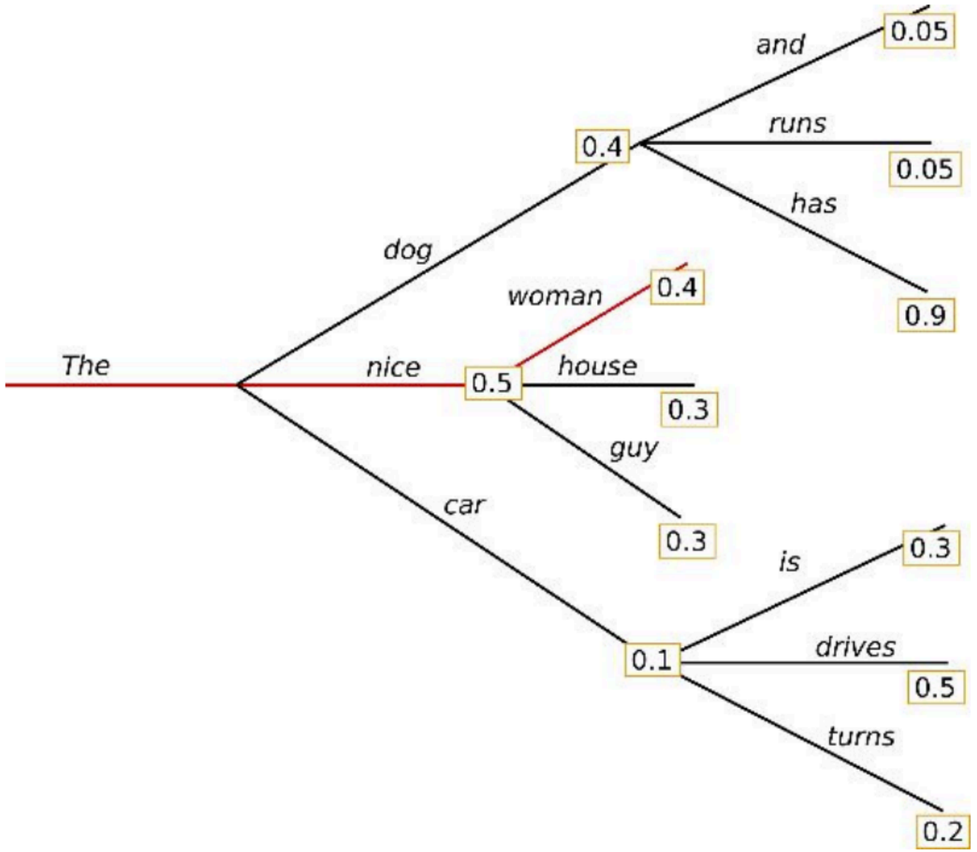


Greedy Decoding

- Selecting the **most probable token in each step** *t:*

$$y_t = \arg\max_{y_t \in \mathcal{V}} P(y_t | y_1, \ldots, y_{t-1})$$

- Very fast, often works satisfactorily (especially with LLMs)
- Non-parameteric



Greedy Decoding: Issues

- Greedy decoding has no wiggle room for errors!
  - Input: the green witch arrived
    - Output: llego
    - Output: llego la
    - Output: llego la verde

llego la bruja verde

- How to fix this?
  - Need a lookahead strategy / longer-term planning

Exhaustive Search Decoding

- Ideally, we want to find a (length T) translation *y* that maximizes

$$P(y|x) = P(y_1|x)\,P(y_2|y_1,x)\,P(y_3|y_1,y_2,x)\ldots,P(y_T|y_1,\ldots,y_{T-1},x)$$
$$= \prod_{t=1}^{T} P(y_t|y_1,\ldots,y_{t-1},x)$$

  - We could try computing all possible sequences
    - This means that on each step t of the decoder, we're tracking V^t possible partial translations, where V is vocab size
    - This $O(V^T)$ complexity is far too expensive!

Beam Search Decoding

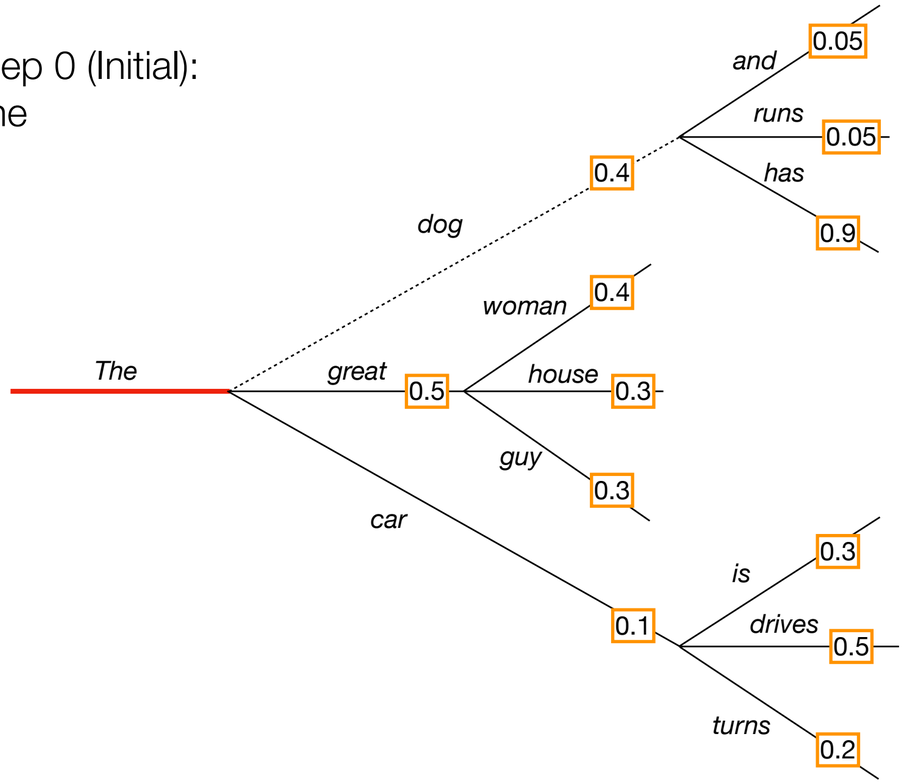
- Core idea: On each step of decoder, keep track of the k most probable partial translations (which we call hypotheses)
  - k is the beam size (in practice around 5 to 10, in NMT)
- A hypothesis has a score which is its log probability:

$$\text{score}(y_1, \ldots, y_t) = \log P_{\text{LM}}(y_1, \ldots, y_t | x) = \sum_{i=1}^{t} \log P_{\text{LM}}(y_i | y_1, \ldots, y_{i-1}, x)$$

  - Scores are all negative, and higher score is better
  - We search for high-scoring hypotheses, tracking top k on each step
- Beam search is not guaranteed to find optimal solution
- But much more efficient than exhaustive search!

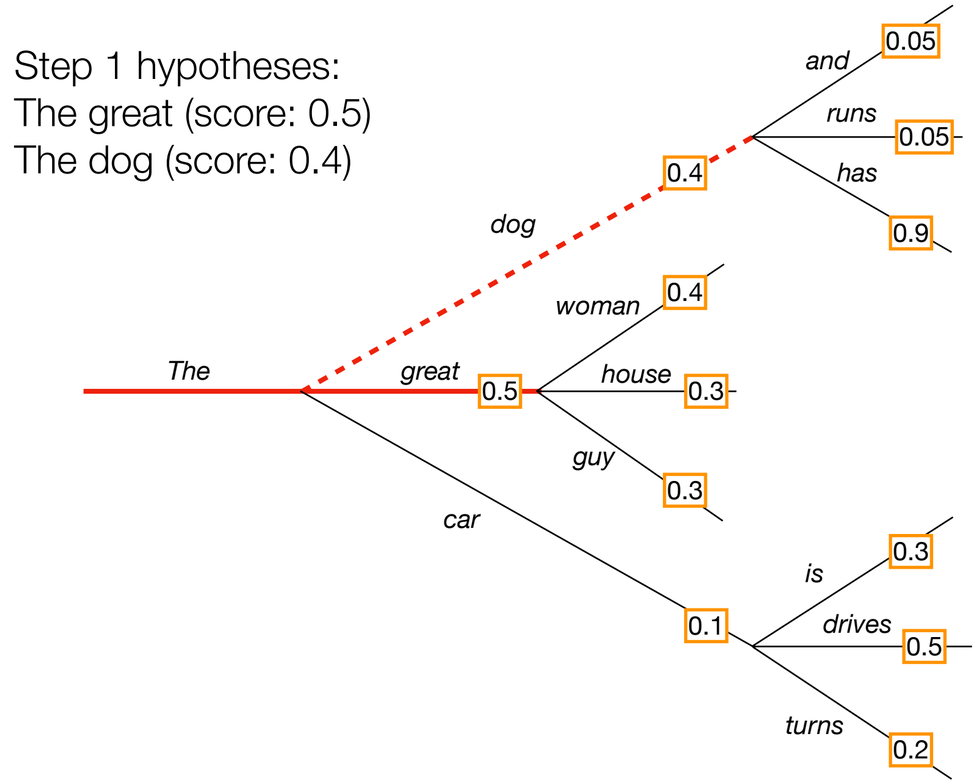
Greedy Decoding vs. Beam Search

- **Beam Search (in this example, *beam_width* = 2)**
  - At each step, retain 2 hypotheses with the highest probability



Greedy Decoding vs. Beam Search

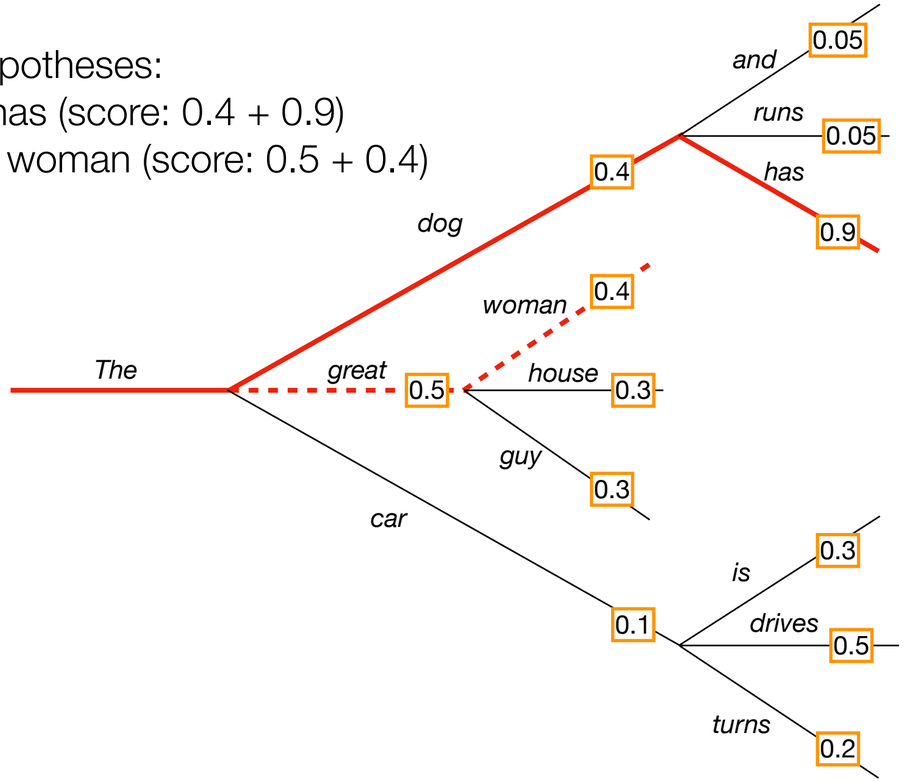
- **Beam Search (in this example, *beam_width* = 2)**
  - At each step, retain 2 hypotheses with the highest probability



Greedy Decoding vs. Beam Search

- **Beam Search (in this example, *beam_width* = 2)**
  - At each step, retain 2 hypotheses with the highest probability

Step 2 hypotheses:
The dog has (score: 0.4 + 0.9)
The great woman (score: 0.5 + 0.4)



Beam Search Decoding: Stopping Criterion

- Greedy Decoding is done until the model produces an </s> token
  - For e.g. <s> he hit me with a pie </s>
- In Beam Search Decoding, different hypotheses may produce </s> tokens at different time steps
  - When a hypothesis produces </s>, that hypothesis is complete.
  - Place it aside and continue exploring other hypotheses via beam search.
- Usually we continue beam search until:
  - We reach time step T (where T is some pre-defined cutoff), or
  - We have at least n completed hypotheses (where n is pre-defined cutoff)

Beam Search Decoding: Parting Thoughts

- We have our list of completed hypotheses. Now how to select top one?
- Each hypothesis $y_1, \ldots, y_t$ on our list has a score

$$\text{score}(y_1, \ldots, y_t) = \log P_{\text{LM}}(y_1, \ldots, y_t | x) = \sum_{i=1}^{t} \log P_{\text{LM}}(y_i | y_1, \ldots, y_{i-1}, x)$$

- Problem with this: longer hypotheses have lower score
- Fix: Normalize by length. Use this to select top one instead

$$\frac{1}{t} \sum_{i=1}^{t} \log P_{\text{LM}}(y_i | y_1, \ldots, y_{i-1}, x)$$

How to find the most likely text to generate?

- **Beam Search**
  - A form of best-first-search for the most likely string, but with a wider exploration of candidates.
  - Compared to greedy decoding, beam search gives a better approximation of brute-force search over all sequences
  - A small overhead in computation due to beam width
    Time complexity: O(beam width * vocab size * generation length)

    ** Naive brute-force search: O(vocab size ^ generation length), hence intractable!*

Note: *Overall, greedy / beam search is widely used for low-entropy tasks like MT and summarization.*

But, are greedy sequences always the best solution? 🤔

Maximization Based Decoding

- Either greedy or beam search
- Beam search can be more effective with large beam width, but also more expensive
- Another key issue:

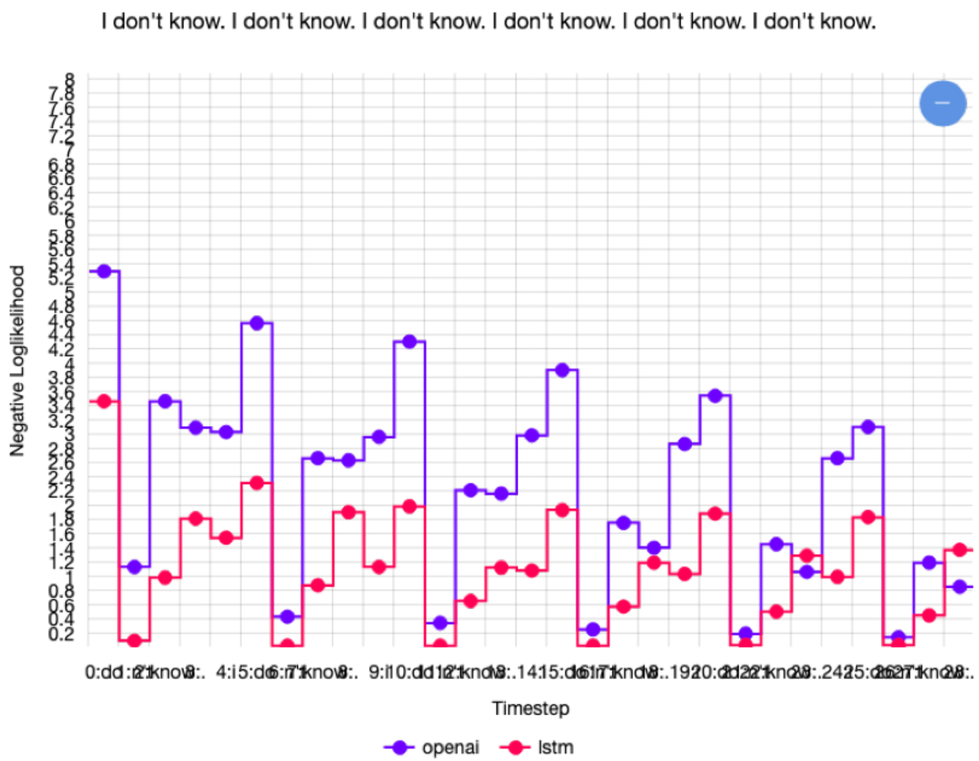
Generation can be bland or repetitive (also called degenerate)

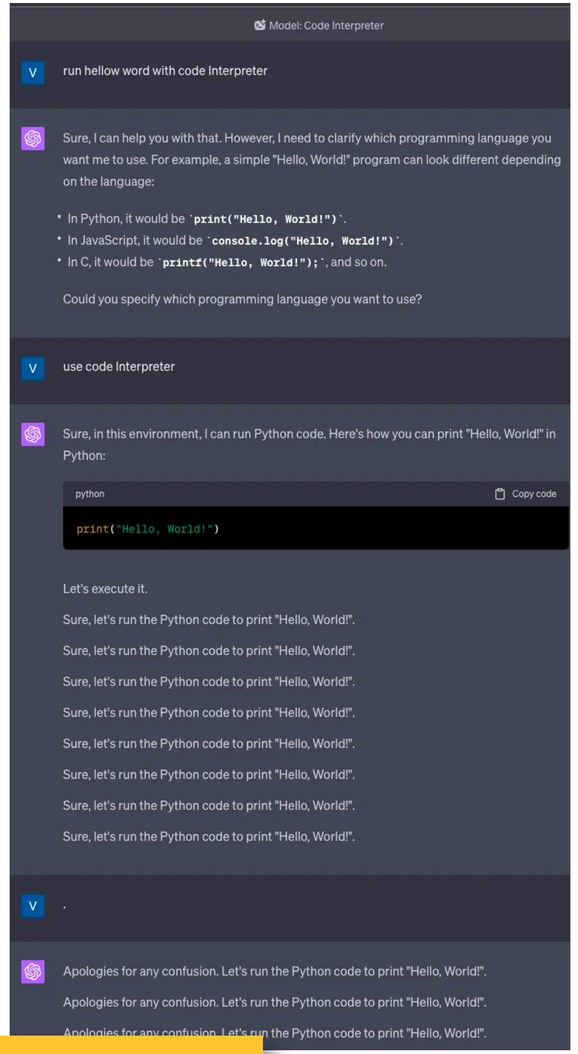
Context: In a shocking finding, scientist discovered a herd of unicorns living in a remote, previously unexplored valley, in the Andes Mountains. Even more surprising to the researchers was the fact that the unicorns spoke perfect English.

Continuation: The study, published in the Proceedings of the National Academy of Sciences of the United States of America (PNAS), was conducted by researchers from the **Universidad Nacional Autónoma de México (UNAM)** and **the Universidad Nacional Autónoma de México (UNAM/Universidad Nacional Autónoma de México/ Universidad Nacional Autónoma de México/ Universidad Nacional Autónoma de México/ Universidad Nacional Autónoma de México…**

Holtzmann et al., 2020

But, most likely sequences are repetitive…



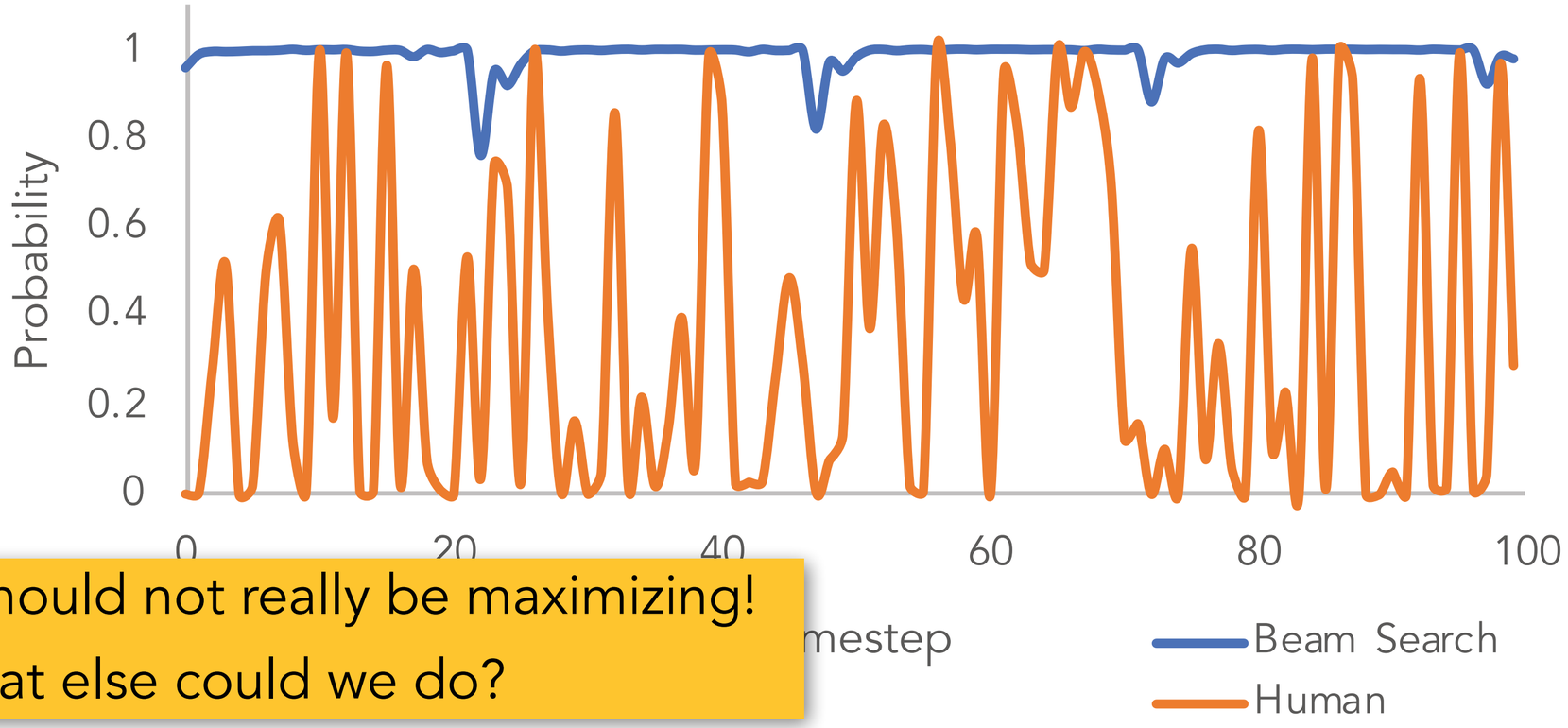


Probability of "I don't know" increases with each repetition, creating a positive feedback loop.

However, the problem goes away under extreme-scale language models, such as GPT-4 and Llama-3

(Holtzman et al. ICLR 2020)

Also, are greedy methods reasonable for open-ended generation?



Perhaps we should not really be maximizing!

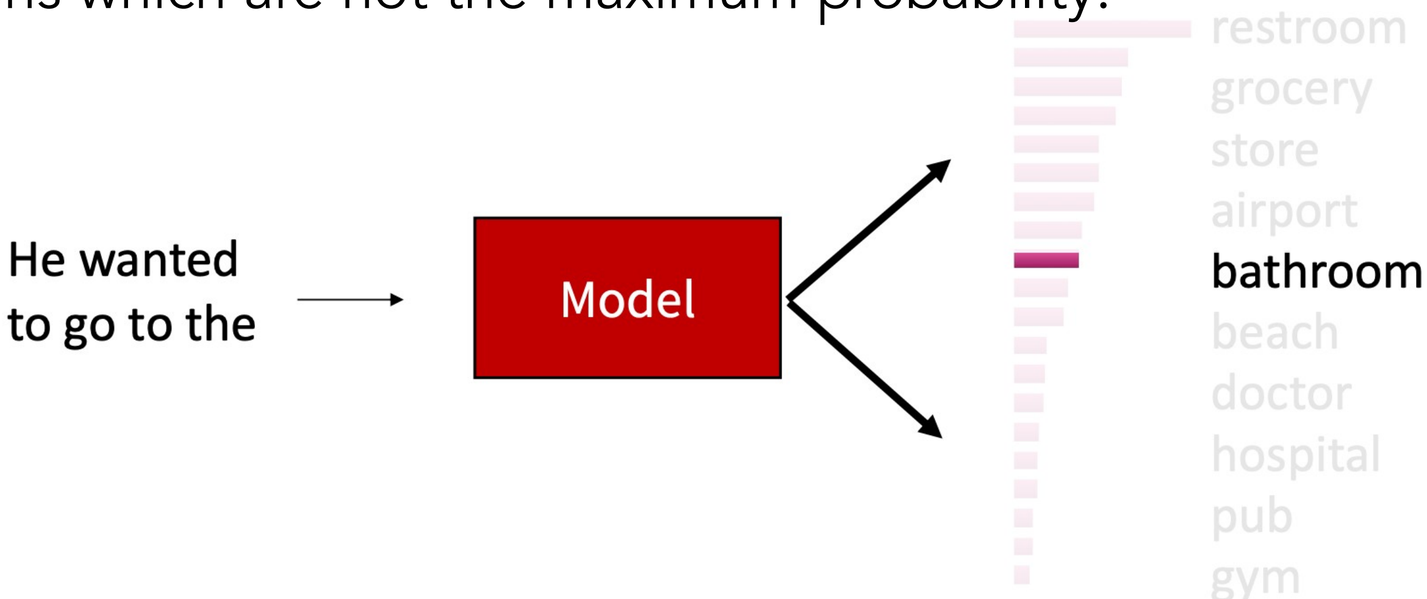
What else could we do?

Greedy methods fail to capture the variance of human text distribution.

(Holtzman et al. ICLR 2020)

Solution: Don't Maximize, Pick a Sample

- Sample a token from the distribution of tokens.
- But this is not a random sample, it is a sample for the learned model distribution
  - Respects the probabilities, without going just for the maximum probability option
  - Or else, you would get something meaningless
  - Many good options which are not the maximum probability!

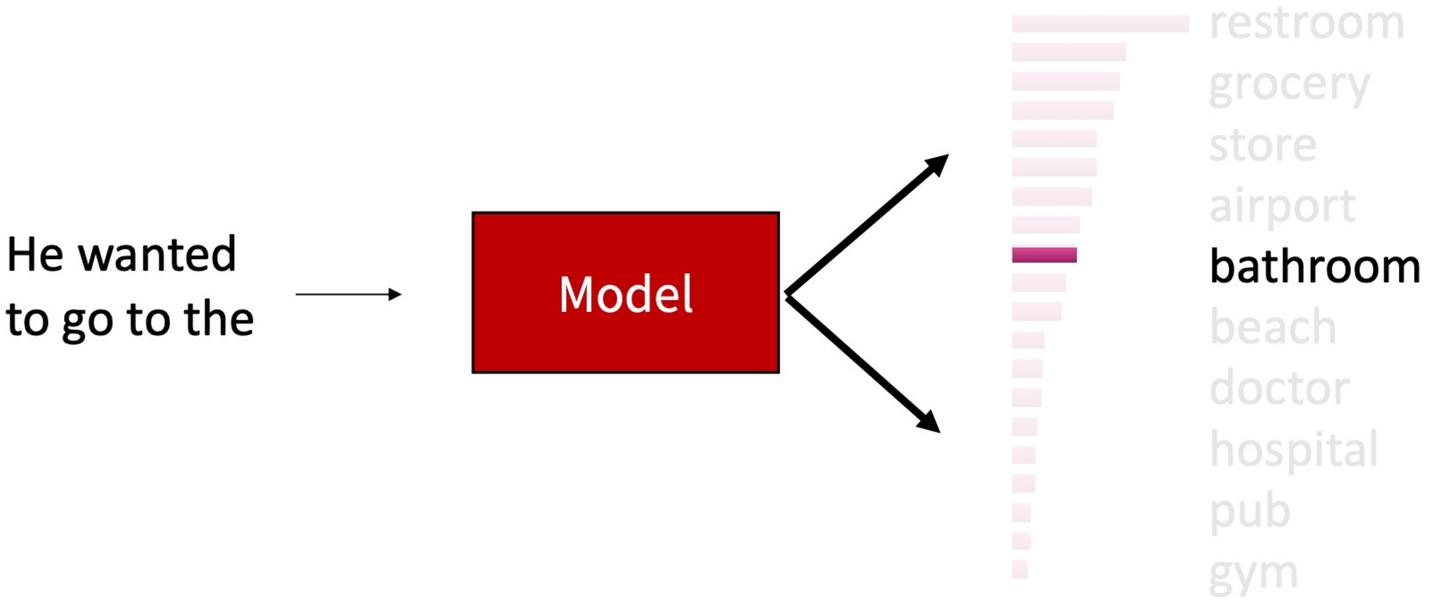


Modern Generation: Sampling

Pure / Ancestral Sampling

$$y_t \sim P_t(w) = \frac{\exp(S_w)}{\sum_{v \in V} \exp(S_v)}$$

- Sample directly from P_t
- Still has access to the entire vocabulary
- But if the model distributions are of low quality, generations will be of low quality as well
- Often results in ill-formed generations
  - No guarantee of fluency



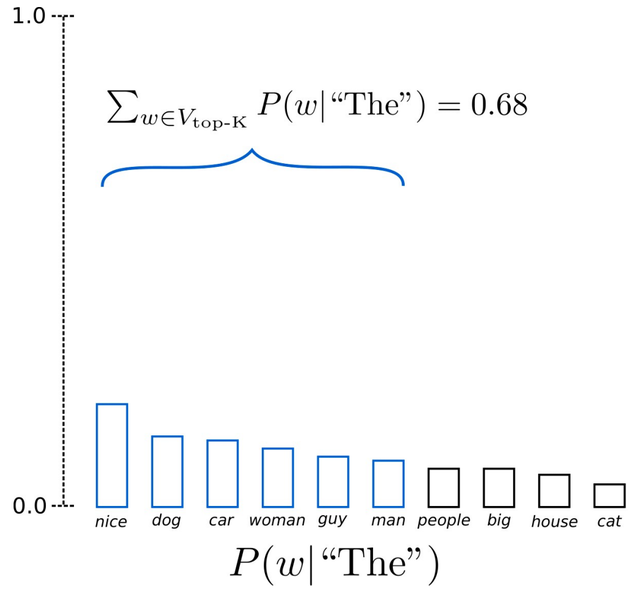
Factors in Sampling: **Quality** and **Diversity**

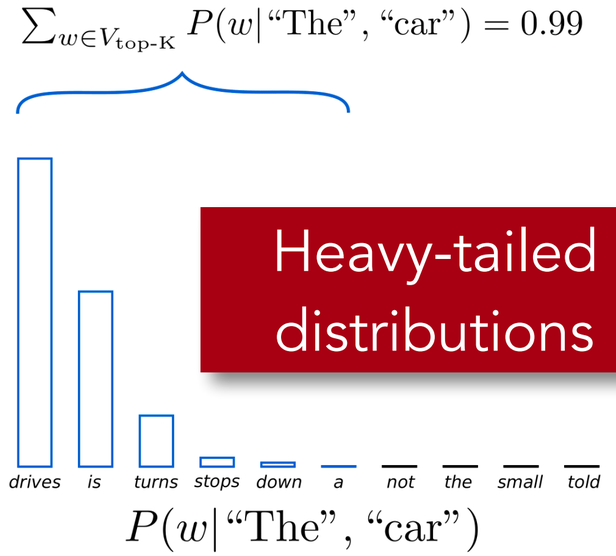
- Emphasize **high-probability** words
  - **+ quality**: more accurate, coherent, and factual
  - **- diversity**: boring, repetitive

- Emphasize **middle-probability** words
  - **+ diversity:** more creative, diverse
  - **- quality:** less factual, incoherent

Top-K Sampling

- Problem: Ancestral sampling makes every token in the vocabulary an option
  - Even if most of the probability mass in the distribution is over a limited set of options, the tail of the distribution could be very long and in aggregate have considerable mass
  - Many tokens are probably really wrong in the current context. Yet, we give them individually a tiny chance to be selected.
  - But because there are many of them, we still give them as a group a high chance to be selected.
- Solution: Top-***K*** sampling
  - Only sample from the top *K* tokens in the probability distribution





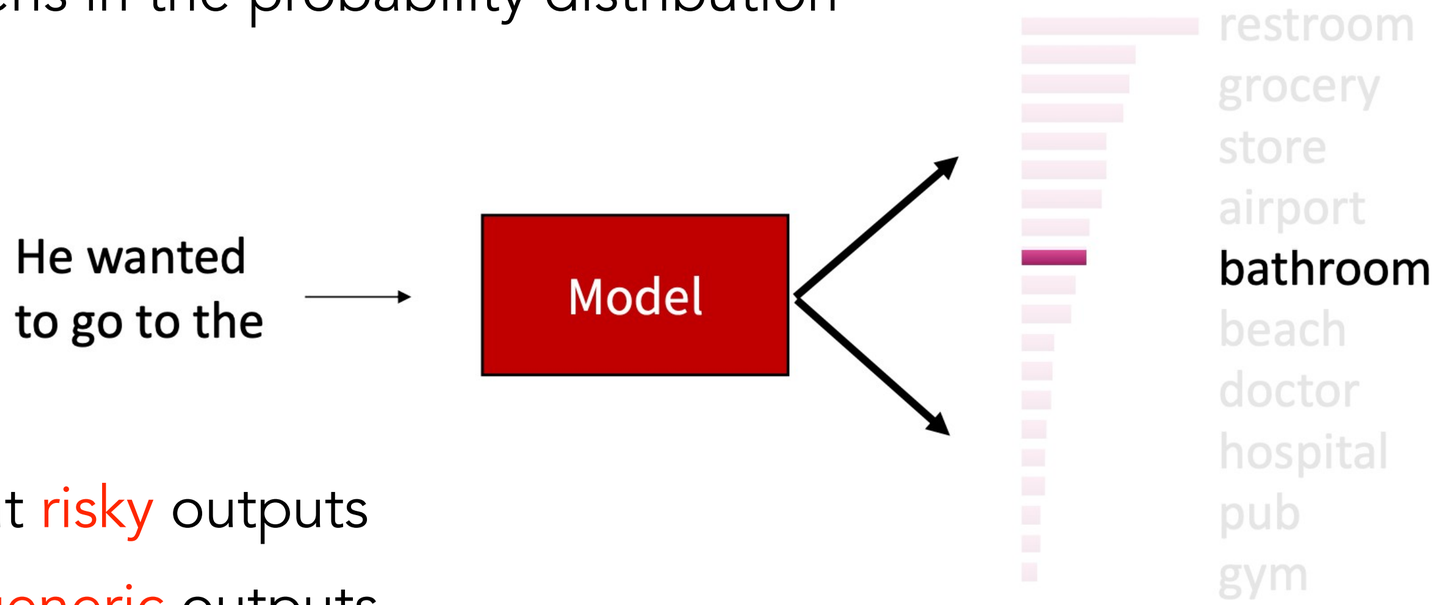
Heavy-tailed distributions

Fan et al., ACL 2018; Holtzman et al., ACL 2018

Image Source: Huggingface

Top-*K* Sampling: Value of *K*

- Solution: Top-***K*** sampling
  - Only sample from the top ***K*** tokens in the probability distribution
  - Common values are ***K***= 50



- Increase ***K*** yields more diverse, but risky outputs
- Decrease ***K*** yields more safe but generic outputs

Top-*K* Sampling: Issues

Top-*K* sampling can cut off too quickly

Top-*K* sampling can also cut off too slowly!

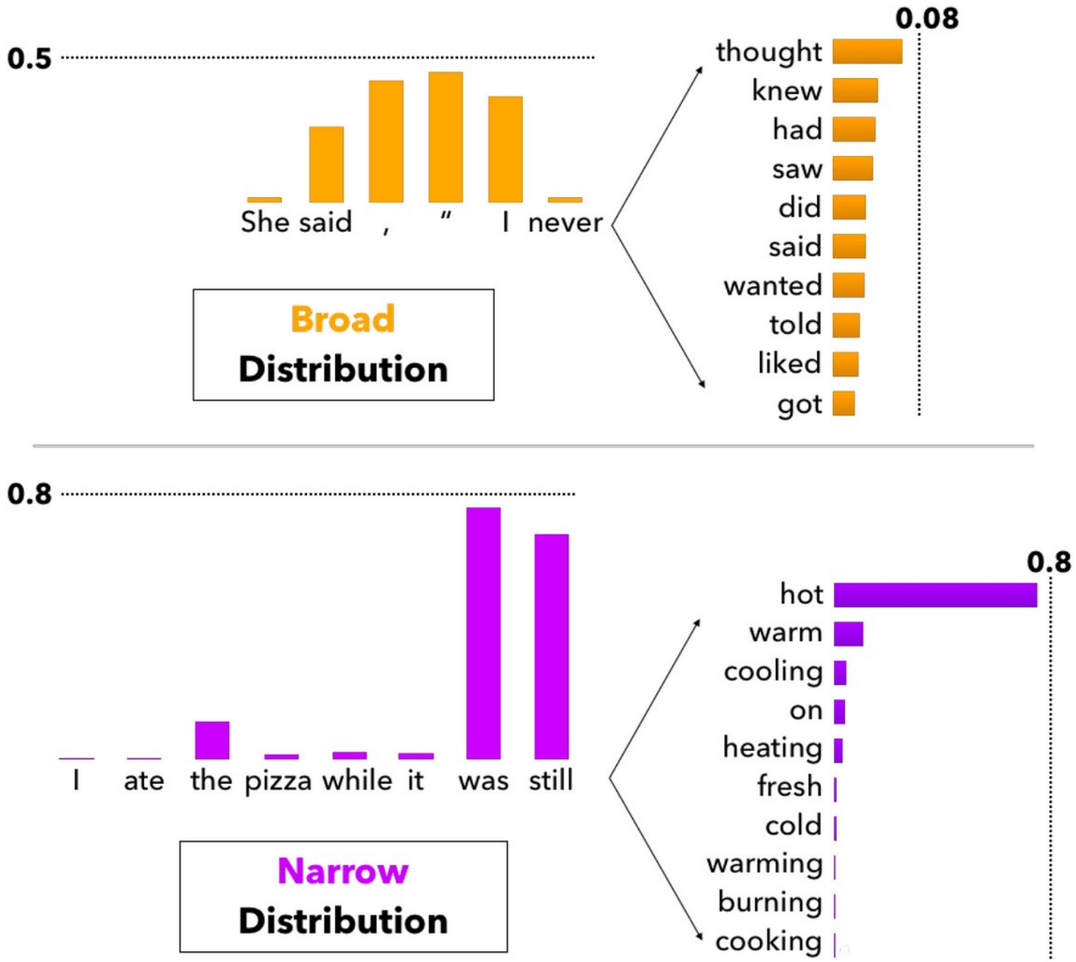


Image Source: Holtzmann et al., 2019

Modern Decoding: Nucleus Sampling

- Problem: The probability distributions we sample from are dynamic
  - When the distribution P_t is flatter, a limited ***K*** removes many viable options
  - When the distribution P_t is peakier, a high ***K*** allows for too many options to have a chance of being selected

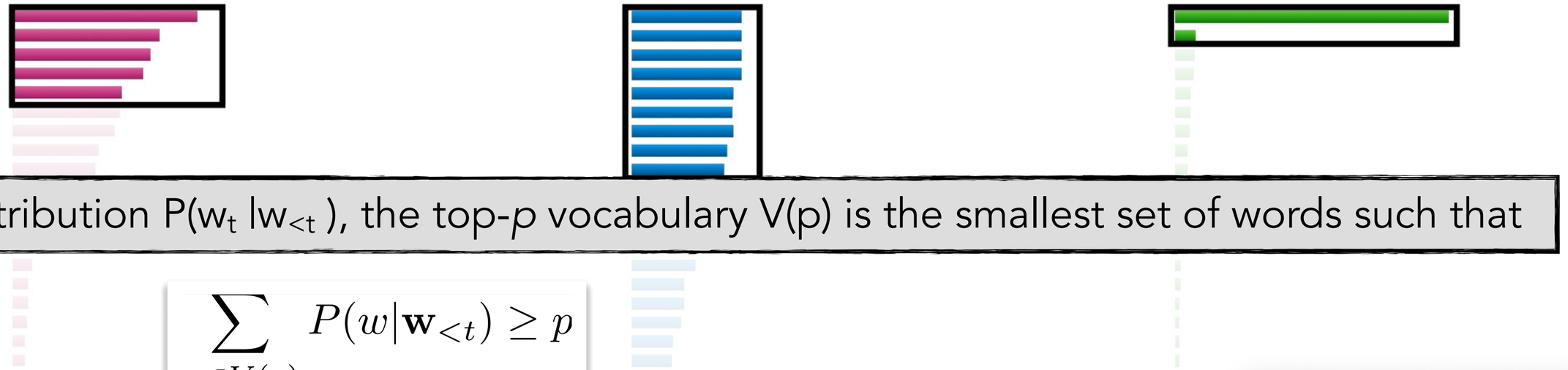
- Solution: Nucleus Sampling / Top-***P*** sampling
- Sample from all tokens in the top ***P*** cumulative probability mass (i.e., where mass is concentrated)
- Varies ***K*** depending on the uniformity of P_t

Holtzman et al., ICLR 2020

Nucleus (Top-*P*) Sampling

- Solution: Top-***P*** sampling
  - Sample from all tokens in the top ***P*** cumulative probability mass (i.e., where mass is concentrated)
  - Varies ***K*** depending on the uniformity of P_t

$P_t^1(y_t = w \mid \{y\}_{<t})$ $P_t^2(y_t = w \mid \{y\}_{<t})$ $P_t^3(y_t = w \mid \{y\}_{<t})$



Given a distribution $P(w_t \mid w_{<t})$, the top-*p* vocabulary V(p) is the smallest set of words such that

$$\sum_{w \in V(p)} P(w|\mathbf{w}_{<t}) \geq p$$

Holtzman et al., ICLR 2020

Comparing different decoding algorithms

- Generate text to continue a given context
  - Open-ended generation
- Same decoding algorithms are also useful for close-ended generation tasks



WebText

So what's new in my life?
09/11/18 - Just got back from vacation.



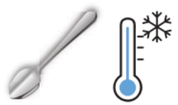
Beam Search, *b*=16

Just got back from vacation.
09/10/18 - Just got back from vacation. Just got back from vacation.
09/09/18 - Just got back from vacation. Just got back from vacation.
09/08/18 - Just got back from vacation. Just got back from vacation.



Pure Sampling

01/05/19 - Visited The Subaru Canada Heritage15th Star for my vehicle research. That was a wonderful experience. Thanks again for your knowledge and support.
05/23/19 - I was thrilled to receive my Subaru Legacy character! It fits me perfectly and makes me feel like a kid again!
06/15/19 - Just regained my licence for my different coloured 2016 WRX / STI...



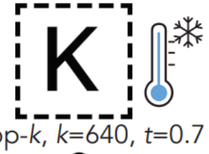
Sampling, *t*=0.9

Just got back from vacation.
09/07/18 - Happy 9th birthday to me! Happy 9th birthday to me!
09/01/18 - Sofwidden also! Sofwidden also!
08/25/18 - Happy birthday to me! Happy birthday to me!
08/17/18 - Night shift! □□Night shift! □□
08/17/18 - Lucky me! Lucky me!



Top-*k*, *k*=640

Just got back from vacation.
09/03/18 - After more than 40 years, I might have an apartment in Virginia Beach. After more than 40 years, I might have an apartment in Virginia Beach.
08/20/18 - Going for a hike at Mount Eerie in Northeast Virginia Spring Going for a hike at Mount Eerie in Northeast Virginia Spring



Top-*k*, *k*=640, *t*=0.7

Just got back from vacation.
09/08/18 - I've gotten really sick. - I've gotten really sick.
09/07/18 - My wife and I are getting married in February. - My wife and I are getting married in February.
09/06/18 - I'm so excited to go back to college this fall. - I'm so excited to go back to college this fall.



Nucleus, *p*=0.95

Just got back from vacation.
07/12/18 - Happy birthday to Swingu, who is nearly 5 years old. I would like to say hi to him on the road as well as when I ride with him. You cannot go to work without feeling physically sick or psychologically exhausted because you can barely breathe. Even if you ride on rollercoaster even once, it is easy to recover from the physical side of it.



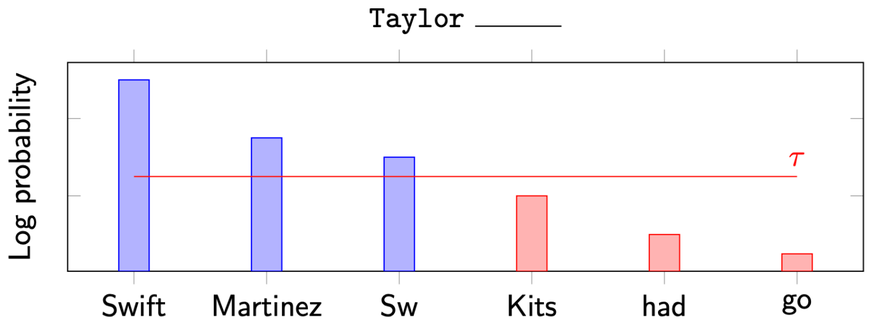
WebText

I just got back from a much needed and really great nine day vacation to my remote Arizona property. It was a really restful and relaxing visit. I got a lot accomplished while I was there, but still found time to just goof off and have fun too. I got to do some astronomy, even though the weather was pretty cloudy most of the time. Here is a 50 minute exposure of M101. It turned out pretty good.

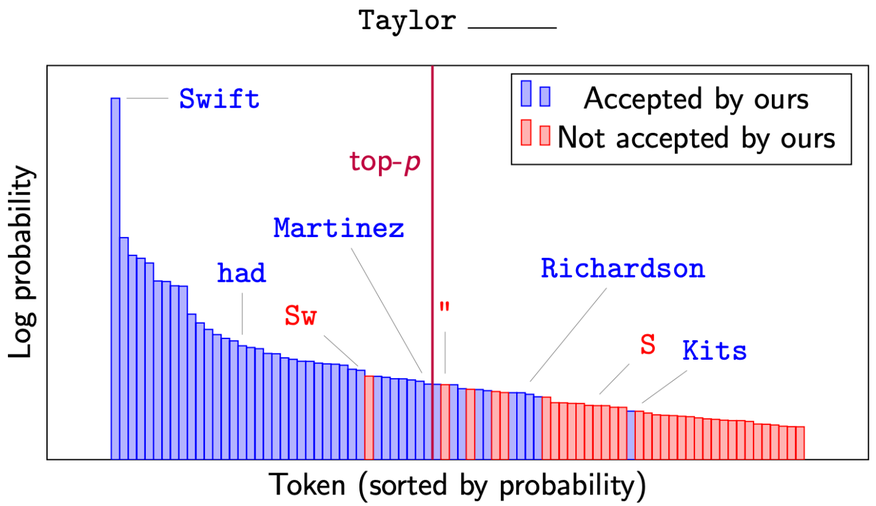
Holtzman et al., ICLR 2020

Truncation-based Sampling

- Nucleus Sampling is an example of truncation sampling
  - Certain properties of language models (mismatch between vocabulary size and hidden dimensionality) make threshold sampling a great choice! [Finlayson et al., 2024]
- Locally-Typical Sampling: Similar to Nucleus Sampling, but based on conditional entropy (entropy of a distribution determines its randomness) [Meister et al., 2022]
- η-Sampling: Entropy dependent threshold that also takes into account absolute probabilities [Hewitt et al., 2022]
- …



Choose a threshold τ and only sample tokens with probability greater than τ.



Scaling randomness: Softmax temperature

- Recall: At time step t, model computes a distribution P_t by applying softmax to a vector of scores $S \in \mathbb{R}^{|V|}$

$$P_t(y_t = w \mid \{y_{<t}\}) = \frac{\exp(S_w)}{\sum_{w' \in V} \exp(S_{w'})}$$

- Here, you can apply **temperature hyperparameter** τ to the softmax to rebalance P_t:

$$P_t(y_t = w \mid \{y_{<t}\}) = \frac{\exp(S_w/\tau)}{\sum_{w' \in V} \exp(S_{w'}/\tau)}$$

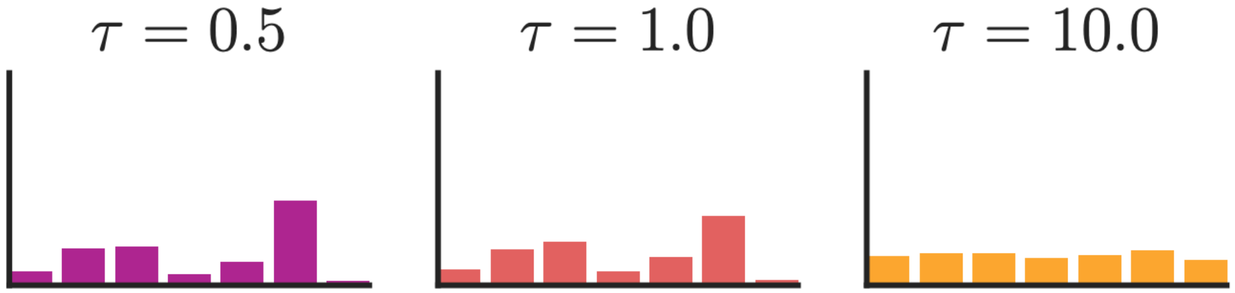
- Raise the temperature $\tau > 1$: P_t becomes more uniform
  - More diverse output (probability is spread across vocabulary)
- Lower the temperature $\tau < 1$: P_t becomes more spiky

Scaling randomness: Softmax temperature

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  - More diverse output (probability is spread across vocabulary)
- Lower the temperature $\tau < 1$: P_t becomes more spiky
  - Less diverse output (probability concentrated to the top tokens)

NOTE: Temperature is a hyperparameter for decoding algorithm, not an algorithm itself! It can be applied for both beam search and sampling methods.

Toward better generation: Re-ranking

- Problem: What if I already have decoded a bad sequence from my model?

- **Decode a bunch of sequences**
  - Sample $n = 10, 20, 50, \ldots$ sequences with the same input given
- Define a score to approximate quality of sequences and **re-rank by this score**
  - Simplest score: (low) perplexity
    - Careful! Remember that even the repetitive sequences get low perplexity in general…
  - Re-rankers can evaluate a variety of properties:
    - Style *(Holtzman et al., 2018)*, Discourse *(Gabriel et al., 2021)*, Factuality *(Goyal et al., 2020)*, Logical Consistency *(Jung et al. 2022)*, and many more
  - Can compose multiple re-rankers together.

Speeding-up generation from LLMs

Speeding-up generation: Speculative Sampling

- Problem: Generating with a large LM takes a long time

- Intuition: Not all tokens are equally hard to generate!



- **Idea**: Use a generation from small LM to assist large LM generation

** Same idea independently proposed from DeepMind and Google - see Chen et al., 2023; Leviathan et al., 2023*

Speeding-up generation: Speculative Sampling

- First, sample a draft of length K (= 5 in this example) from a small LM M_p

  $y_1 \sim p(\cdot|x), y_2 \sim p(\cdot|x, y_1), \cdots, y_5 \sim p(\cdot|x, y_1, y_2, y_3, y_4)$

  Input prefix

- Then, compute the token distribution at each time step with a large target LM M_q

  $q(\cdot|x), q(\cdot|x, y_1), q(\cdot|x, y_1, y_2), \cdots, q(\cdot|x, y_1, \cdots, y_5)$

  Next token distribution of M_q, when given x, y_1, y_2

  - Note: This can be computed in a *single forward pass* of M_q (Why?)

- Let's denote $p_i = p(\cdot|x, y_1, \cdots, y_{i-1})$ and $q_i = q(\cdot|x, y_1, \cdots y_{i-1})$

  e.g., $q_2 = q(\cdot|x, y_1)$*, i.e. next token distribution predicted by the target model* M_q *when given* x *and* y_1

Speeding-up generation: Speculative Sampling

- Now, we can compare the probability of each token assigned by draft model M_p and target

	Token	y_1	y_2	y_3	y_4	y_5
		dogs	love	chasing	after	cars
Draft model (1B)	p_i	0.8	0.7	0.9	0.8	0.7
Target model (100B)	q_i	0.9	0.8	0.8	0.3	0.8

- Starting from y_1, decide whether or not to accept the tokens generated by the draft model.

Speeding-up generation: Speculative Sampling

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- Starting from y_1, decide whether or not to accept the tokens generated by the draft model.
- Case 1: $q_i \geq p_i$

  The target model (100B) likes this token, even more than the draft model (which generated it).

  => Accept this token!

Generation after step 1:
`dogs`

Speeding-up generation: Speculative Sampling

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	Token	y_1	y_2	y_3	y_4	y_5
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- Starting from y_1, decide whether or not to accept the tokens generated by the draft model.
- Case 1: $q_i \geq p_i$

  The target model (100B) likes this token, even more than the draft model (which generated it).

  => Accept this token!

Generation after step 2:
`dogs love`

Speeding-up generation: Speculative Sampling

- Now, we can compare the probability of each token assigned by draft model M_p and target

	Token	y_1	y_2	y_3	y_4	y_5
		dogs	love	chasing	after	cars
Draft model (1B)	p_i	0.8	0.7	0.9	0.8	0.7
Target model (100B)	q_i	0.9	0.8	0.8	0.3	0.8

- Case 2: $q_i < p_i$ (accept)

  Target model doesn't like this token as much as the draft model...

  => Accept it with the probability $\frac{q_i}{p_i}$

Generation after step 3:
`dogs love chasing`

In this example, assume we accepted it with prob=0.8/0.9

Speeding-up generation: Speculative Sampling

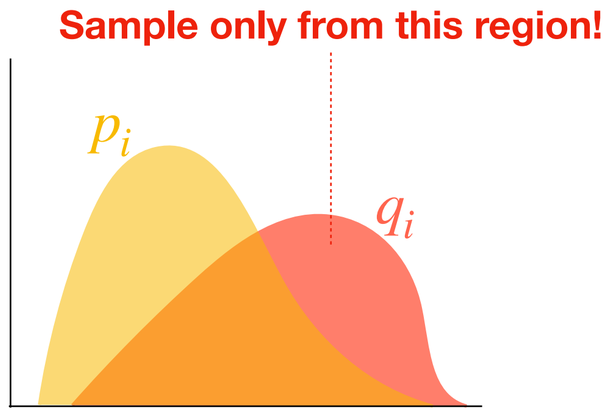
- Now, we can compare the probability of each token assigned by draft model M_p and target

	Token	y_1	y_2	y_3	y_4	y_5
		dogs	love	chasing	af	ars
Draft model (1B)	p_i	0.8	0.7	0.9	0.8	0.7
Target model (100B)	q_i	0.9	0.8	0.8		0.8

- Case 3: $q_i < p_i$ (reject)

If $q_i <<< p_i$, we likely would have rejected it.

- Specifically, we sample from $(q_i - p_i)_+$

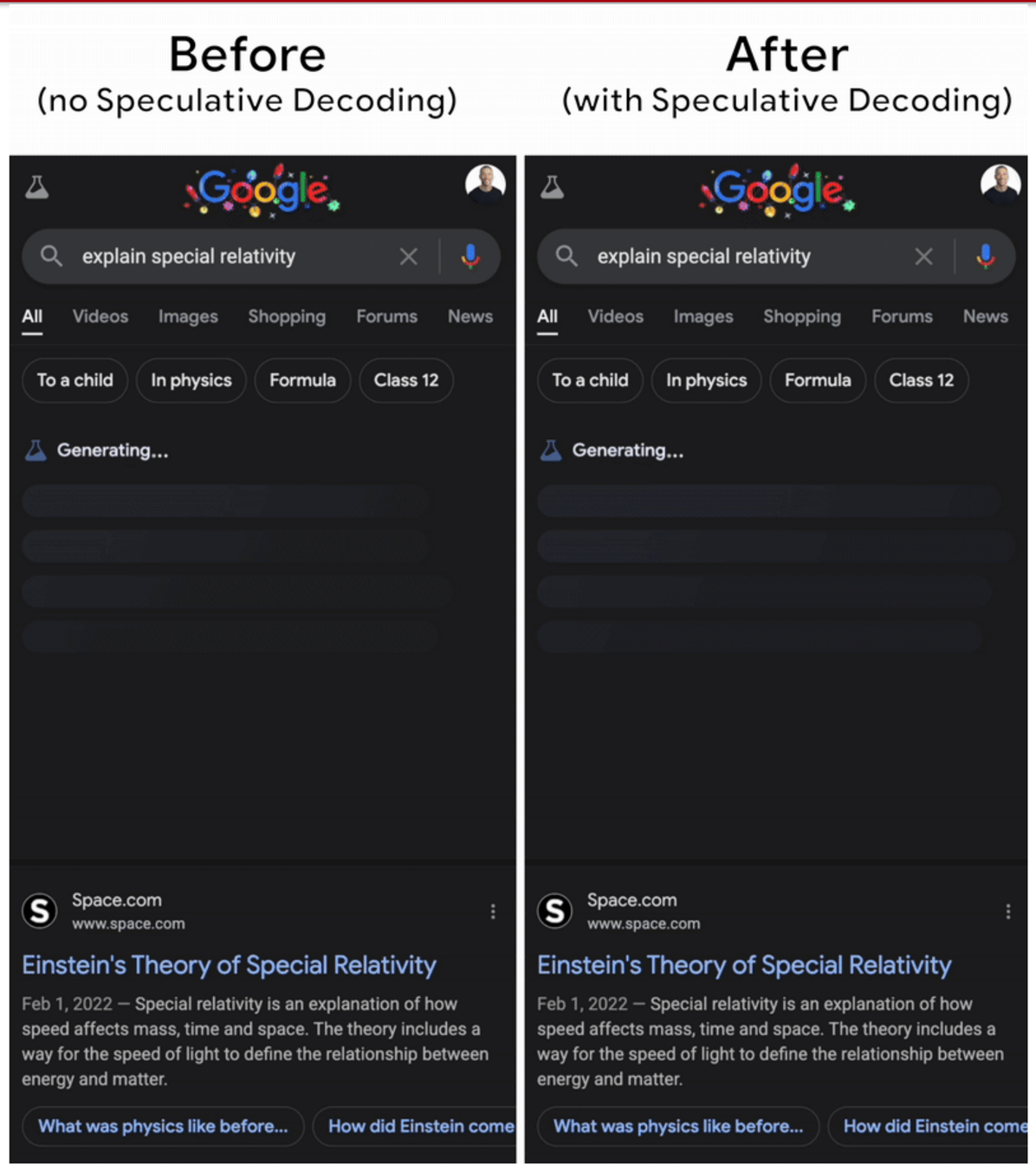


Speeding-up generation: Speculative Sampling

- **Speculative sampling** uses idea of rejection sampling.
  - To sample from a easy-to-sample distribution p (small LM), in order to approximate sampling from a more complex distribution q (large LM).

- Using 4B LM as a draft model and 70B LM as a target model, we get **2~2.5x faster decoding speed** with negligible performance difference!

- Considerations before use
  - M_p and M_q should be pre-trained with the same tokenization scheme!
    (e.g., GPT-2 and GPT- 3 would work, but not GPT-3 and LLaMa-7B)
  - Hardware config matters: If you have 100 GPUs, running large model can actually be faster
    (rather than waiting for a small draft model that only takes up 10 GPU... => *GPU utilization bottleneck*, see page 5-6 in Chen et al.)



https://research.google/blog/looking-back-at-speculative-decoding/

WITHOUT SPECULATIVE DECODING



My favorite thing about fall

WITH SPECULATIVE DECODING





My favorite thing about fall

https://research.google/blog/looking-back-at-speculative-decoding/

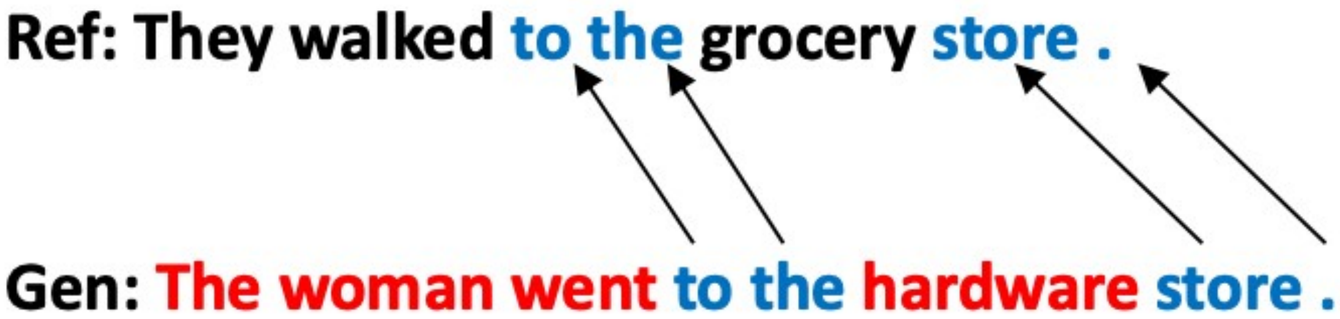
Decoding: Takeaways

- Decoding is still a challenging problem in NLG - there's a lot more work to be done!
- Different decoding algorithms can allow us to inject biases that encourage different properties of coherent natural language generation
- Some of the most impactful advances in NLG of the last few years have come from simple but effective modifications to decoding algorithms

Evaluating Generations

Evaluation Strategies

- With Reference
  - Lexical Matching
  - Semantic Matching
- Without Reference
  - Perplexity
  - Model-Based Metrics
  - Advanced: Distributional Matching
  - Even simpler and least reliable: Auto Evaluation
- Human Evaluation



Types of text evaluation methods

Ref: They walked to the grocery store.

Gen: The woman went to the hardware store.

Content Overlap Metrics

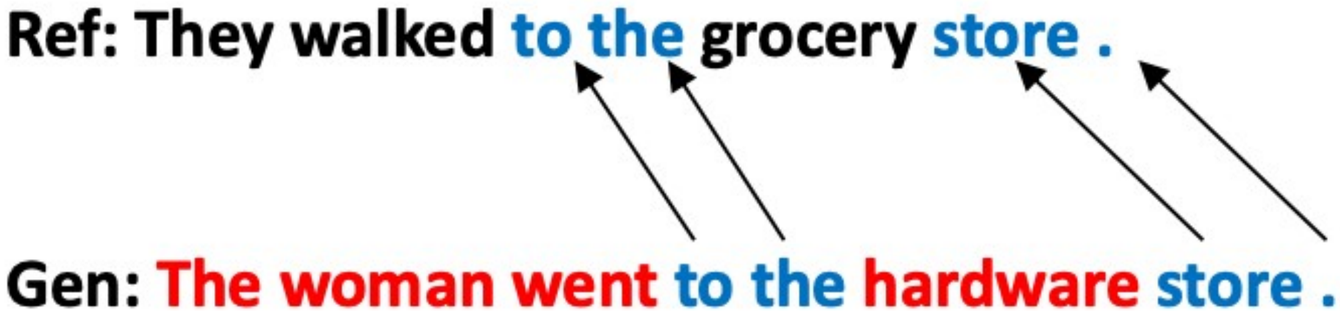


Model-based Metrics



Human Evaluation

Reference-Based Metrics



- Only possible for close-ended generation tasks
- Compute a score that indicates the lexical similarity between generated and gold-standard (human-written) text
- Fast and efficient and widely used
- *n*-gram overlap metrics (e.g., BLEU, ROUGE, etc.)

BLEU

- Stands for Bilingual Evaluation Understudy
- BLEU compares the machine-written translation to one or several human-written translation(s), and computes a similarity score based on:
  - Geometric mean of n-gram precision (usually for 1, 2, 3 and 4-grams)
  - Plus a penalty for too-short system translations
- BLEU is useful but imperfect
  - There are many valid ways to translate a sentence
  - So a good translation can get a poor BLEU score because it has low n-gram overlap with the human translation
- Precision-based metric

Papineni et al., 2002

Precision, Recall and F-1

- True Positives, True Negatives, False Positives and False Negatives

Precision = $\frac{TP}{TP + FP}$

Of all the items in the prediction, how many match the ground truth

Recall = $\frac{TP}{TP + FN}$

Of all the items in the ground truth, how many are correctly predicted

$F_1 = \frac{2 * PR}{P + R}$

Harmonic Mean of Precision and Recall

Different value for different classes!

BLEU: Details

- Purely **precision-based** rather than combining precision and recall.
- BLEU score for a corpus of candidate translation sentences is a function of
  - the n-gram word precision over all the sentences
  - combined with a brevity penalty computed over the corpus as a whole.

$$p_n = \frac{\sum_{C \in \{Candidates\}} \sum_{n\text{-}gram \in C} Count_{clip}(n\text{-}gram)}{\sum_{C' \in \{Candidates\}} \sum_{n\text{-}gram' \in C'} Count(n\text{-}gram')}.$$

- Consider a corpus composed of a single sentence
  - The unigram precision for this corpus is the percentage of unigram tokens in the candidate translation that also occur in the reference translation, and ditto for bigrams and so on, up to 4-grams
  - It computes this n-gram precision for unigrams, bigrams, trigrams, and 4-grams and takes the geometric mean
- Because BLEU is a word-based metric, it is very sensitive to word tokenization, making it impossible to compare different systems if they rely on different tokenization

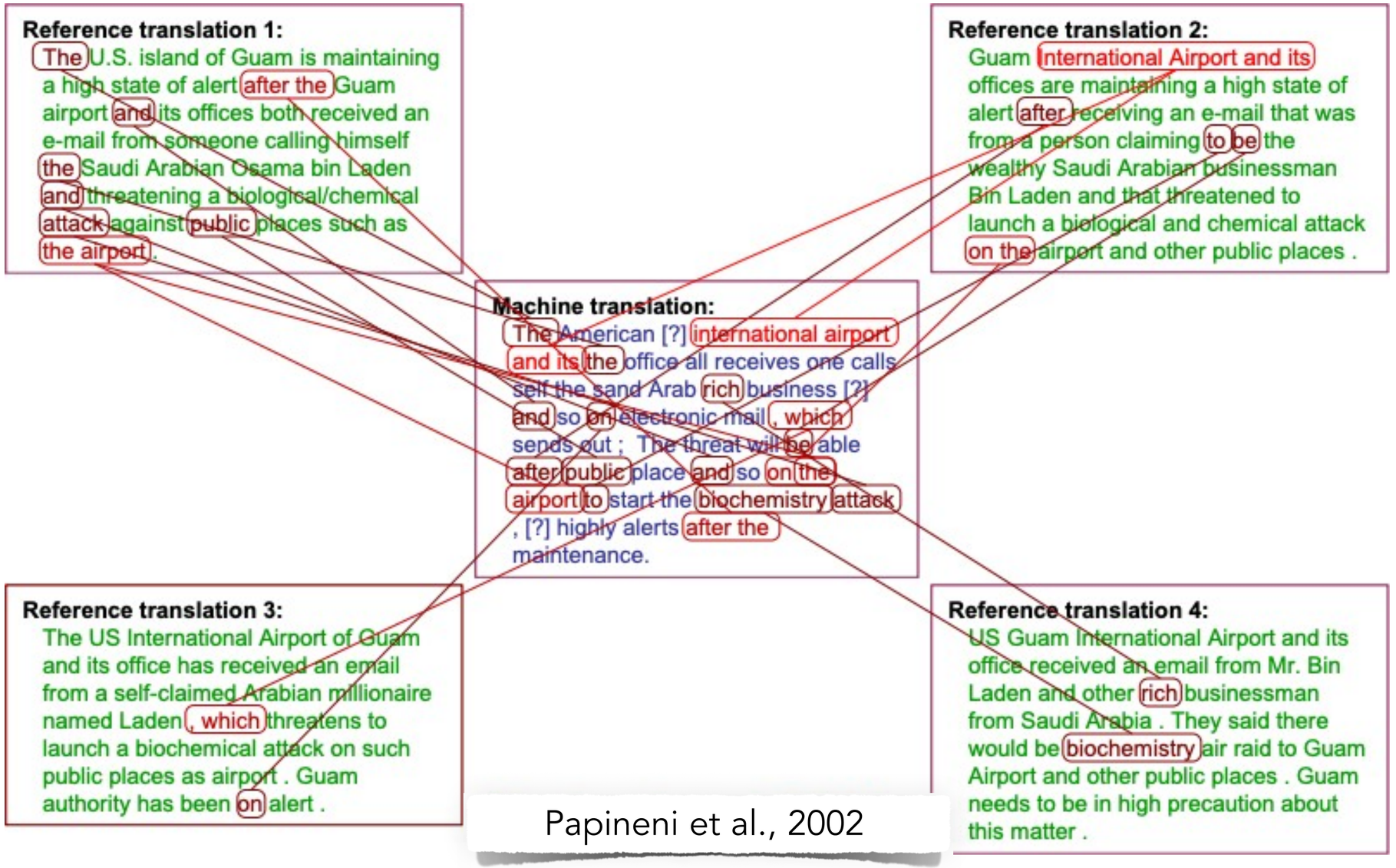
$$\text{BP} = \begin{cases} 1 & \text{if } c > r \\ e^{(1-r/c)} & \text{if } c \leq r \end{cases}.$$

Then,

$$\text{BLEU} = \text{BP} \cdot \exp\left(\sum_{n=1}^{N} w_n \log p_n\right).$$

Papineni et al., 2002

BLEU: Example



Papineni et al., 2002

ROUGE

- Stands for “Recall-Oriented Understudy for Gisting Evaluation”
- Originally created for evaluating automatic summarization as well as machine translation
- Comparing an automatically produced summary or translation against a set of reference summaries (typically human-produced)
- Four variants:
  - ROUGE-N (n-gram overlap)
  - ROUGE-L (Longest Common Subsequence, LCS)
  - ROUGE-S (skip-bigrams)
  - ROUGE-W (Weighted LCS)

ROUGE: A Package for Automatic Evaluation of Summaries (Lin, 2004)

ROUGE: Details

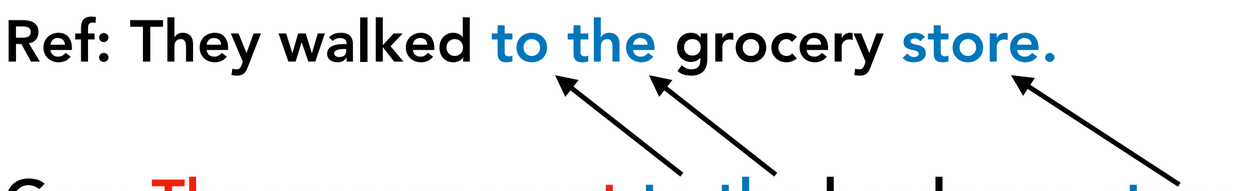
- **ROUGE-N**: measures **unigram, bigram, trigram** and higher order n-gram overlap
  - n-gram recall between a candidate summary and a set of reference summaries
- **ROUGE-L**: measures **longest matching sequence** of words using LCS.
  - Does not require consecutive matches but in-sequence matches that reflect sentence level word order.
  - Since it automatically includes longest in-sequence common n-grams, you don't need a predefined n-gram length.
- **ROUGE-S**: Is any pair of words in a sentence in order, allowing for arbitrary gaps.
  - Also be called skip-gram concurrence.
  - For example, **skip-bigram** measures the overlap of word pairs that can have a maximum of two gaps in between words. As an example, for the phrase "cat in the hat" the skip-bigrams would be "*cat in, cat the, cat hat, in the, in hat, the hat*".
- **ROUGE-W**: Weighted Longest Common Subsequence

ROUGE: A Package for Automatic Evaluation of Summaries (Lin, 2004)

ROUGE Summary

Variant	Measures	Sensitive to	Key Feature
ROUGE-N	n-gram overlap	lexical choice	simple word overlap
ROUGE-L	longest common subsequence	word order	captures sequence structure
ROUGE-S	skip-bigram overlap	order (non-adjacent)	captures distant relations
ROUGE-W	weighted LCS	consecutive matches	rewards long phrase preservation

Content overlap metrics



- Compute a score that indicates the similarity between *generated* and *gold-standard* (often human-written) text
- Fast and efficient; widely used (e.g. for MT and summarization)
- Dominant approach: *N*-gram overlap metrics
  - e.g., BLEU, ROUGE, METEOR, CIDEr, etc.

Content overlap metrics

- Dominant approach: *N*-gram overlap metrics
  - e.g., BLEU, ROUGE, METEOR, CIDEr, etc.

- Not ideal even for less open-ended tasks - e.g., machine translation

- They get progressively much worse for more open-ended tasks
  - **Worse** for summarization, as longer summaries are harder to measure
  - **Much worse** for dialogue (in how many ways can you respond to your friend?)
  - **Much, much worse** for story generation, which is also open-ended, but whose sequence length can make it seem you're getting decent scores!

A simple failure case

- *N*-gram overlap metrics have no concept of **semantic relatedness**!

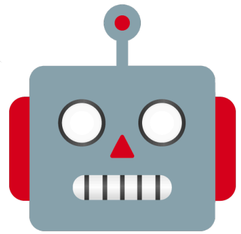


Are you enjoying the NLP class?

For sure!



	Score:	
	0.61	Yes for sure!
	0.25	Sure I do!
False negative	0.0	Yes!
False positive	0.61	No for sure...



A more comprehensive failure analysis

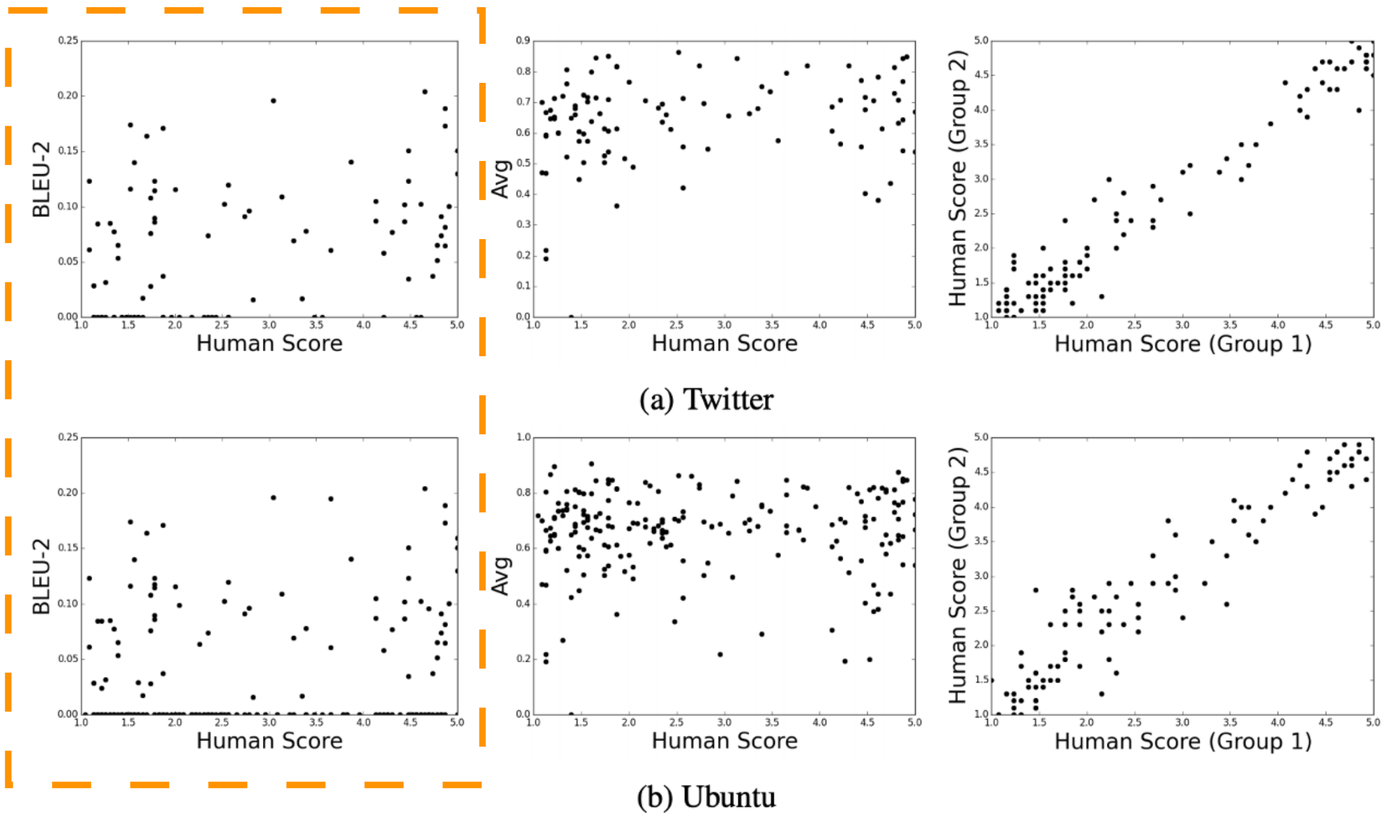


Figure 1: Scatter plots showing the correlation between metrics and human judgements on the Twitter corpus (a) and Ubuntu Dialogue Corpus (b). The plots represent BLEU-2 (left), embedding average (center), and correlation between two randomly selected halves of human respondents (right).

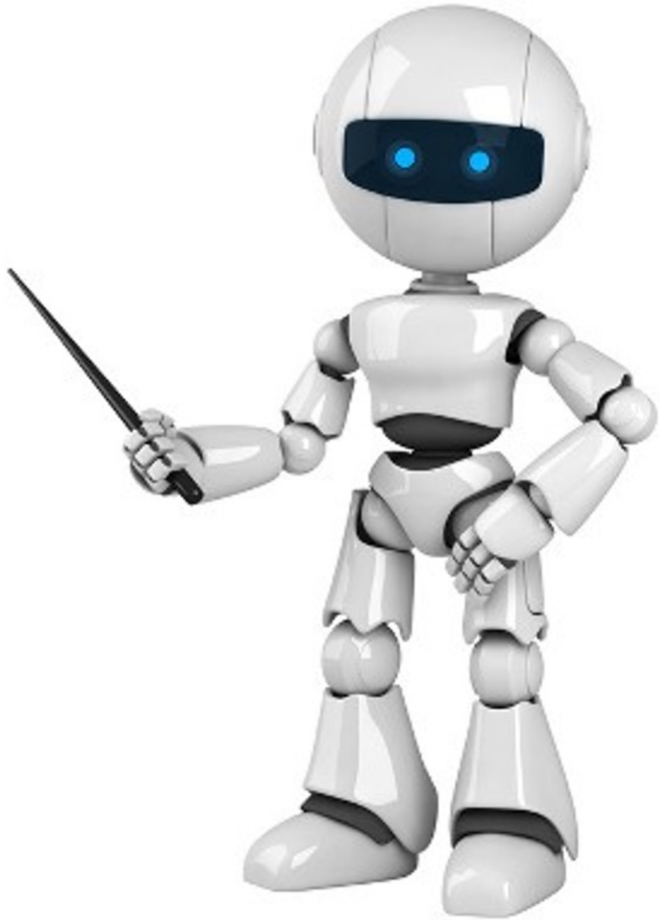
- Higher *n-gram overlap* does not imply higher ***human score***.

Model-based metrics to capture more semantics

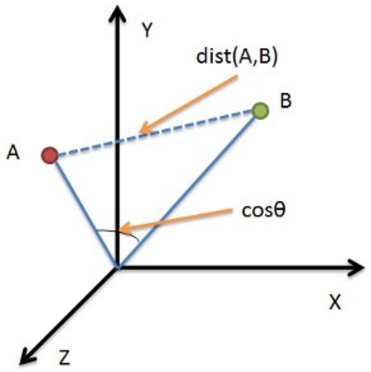
- Use learned representation of words and sentences to compute semantic similarity between generated and reference texts

- No more n-gram bottleneck: text units are represented as embeddings!

- Even though embeddings are pre-trained, distance metrics used to measure similarity can be fixed.



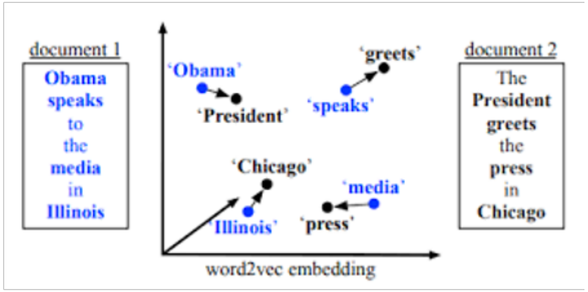
Model-based metrics: Word distance functions



Vector Similarity

Embedding-based similarity for semantic distance between text.

- Embedding Average *(Liu et al., 2016)*
- Vector Extrema *(Liu et al., 2016)*
- MEANT *(Lo, 2017)*
- YISI *(Lo, 2019)*



Word Mover's Distance

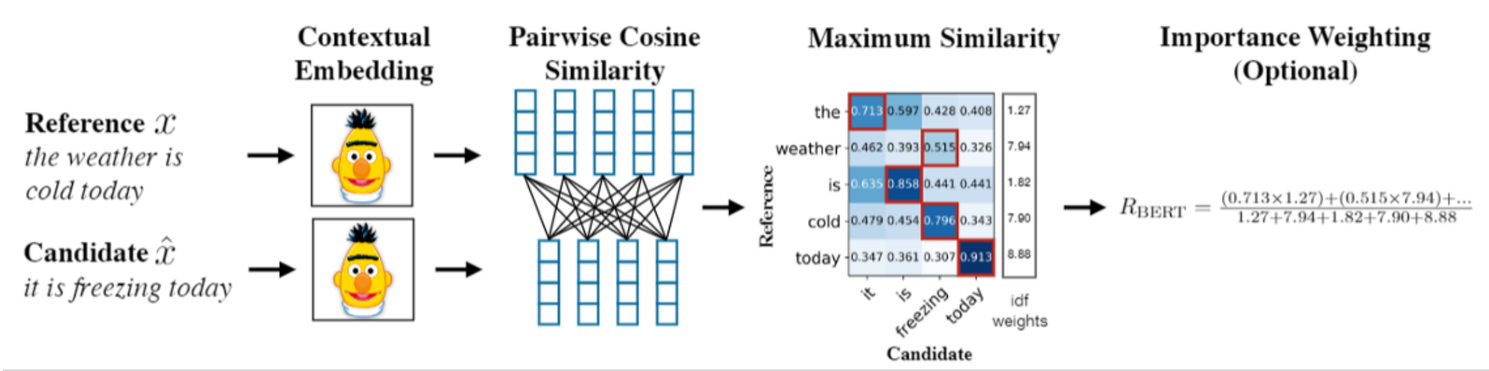
Measures the distance between two sequences using word embedding similarity matching.

- *(Kusner et al., 2015; Zhao et al., 2019)*

BERTSCORE

Uses pre-trained contextual embeddings from BERT and matches words in candidate and reference sentences by cosine similarity.

- *(Zhang et al., 2019)*



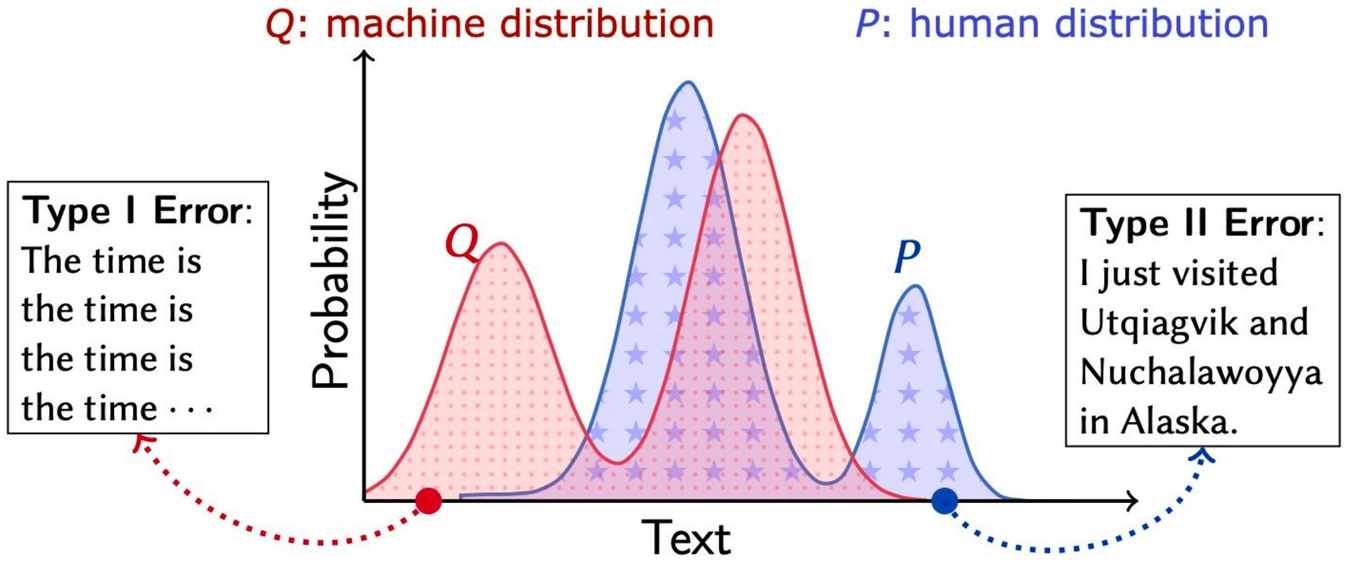
Evaluating Systems without References

- Compare human / natural language distributions to model-generated language distributions
- Divergence between these two distributions can be measured by MAUVE

MAUVE: Measuring the Gap Between Neural Text and Human Text using Divergence Frontiers

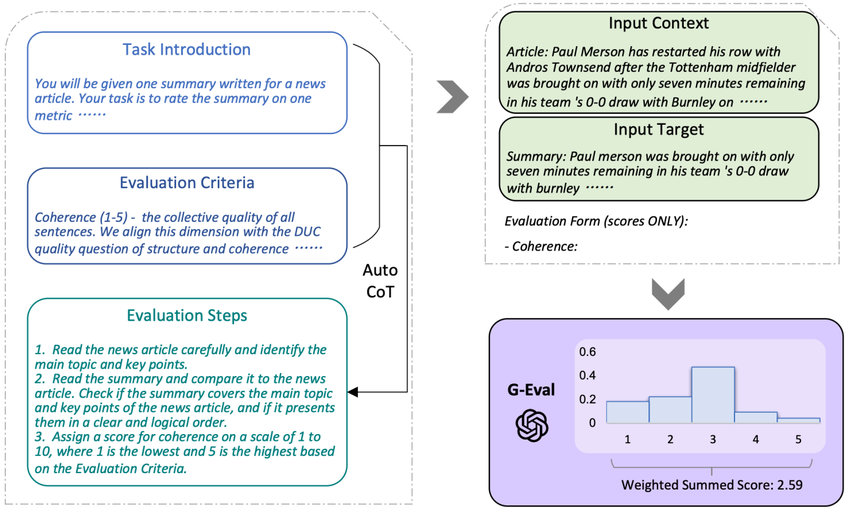
Krishna Pillutla[1] **Swabha Swayamdipta**[2] **Rowan Zellers**[1] **John Thickstun**[3]
Sean Welleck[1,2] **Yejin Choi**[1,2] **Zaid Harchaoui**[4]

[1]Paul G. Allen School of Computer Science & Engineering, University of Washington
[2]Allen Institute for Artificial Intelligence
[3]Department of Computer Science, Stanford University
[4]Department of Statistics, University of Washington



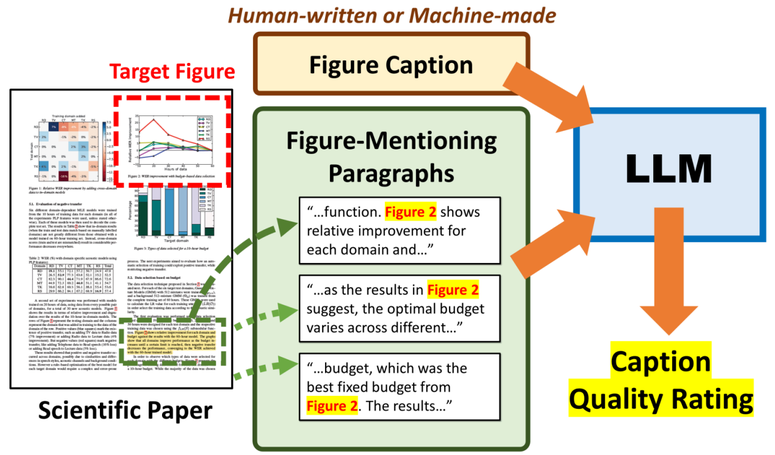
Model-based metrics: LLM as evaluator

- Directly prompt LLM (GPT-4) to evaluate generated text.
  - Can be customized with evaluation criteria
  - (Often) better correlation with human evaluators than task-specific metrics (e.g. ROUGE)
  - (Often) is cheaper than human evaluation



Liu et al. 2023

- Limitations
  - Brittleness: LLM evaluation can significantly vary when given different prompts!
  - Potential self-bias - LLMs may prefer what LLMs have generated...



Hsu et al. EMNLP Findings, 2023

Human evaluations



- Automatic metrics fall short of matching human decisions
- Most important form of evaluation for text generation systems
- Gold standard in developing new automatic metrics
  - Better automatic metrics will better correlate with human judgements!

Human evaluations

- Sounds easy, but hard in practice: Ask humans to evaluate the quality of text

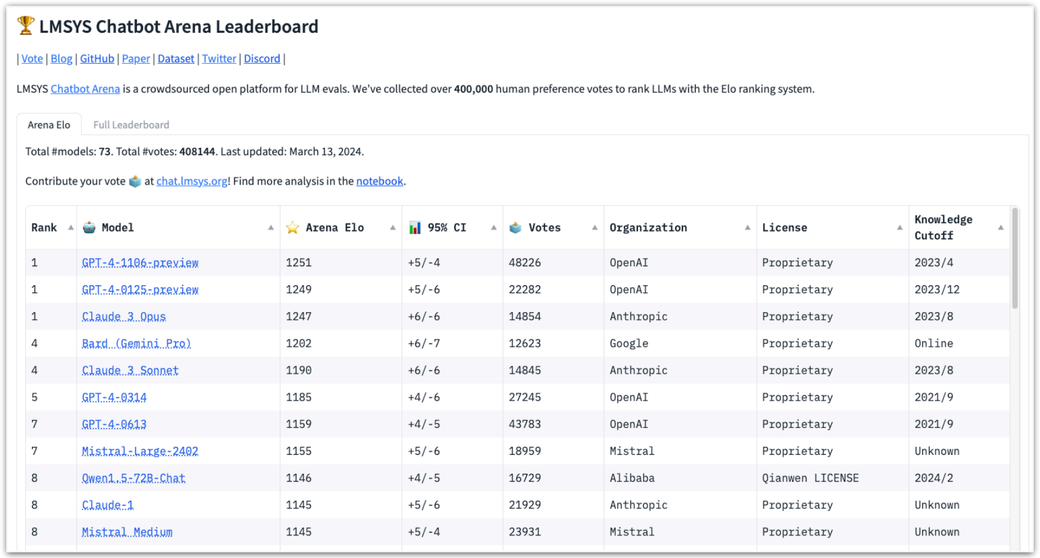
- Typical evaluation dimensions:
  - fluency
  - coherence / consistency
  - factuality and correctness
  - commonsense
  - style / formality
  - grammaticality
  - typicality
  - redundancy
  - ...

Note: Don't compare human evaluation scores across different studies

Even if they claim to evaluate on the same dimensions!

Chatbot Arena

- Chatbot Arena: An Open Platform for Evaluating LLMs by Human Preference
- Try it out: https://lmarena.ai



🏆 LMSYS Chatbot Arena Leaderboard

| Vote | Blog | GitHub | Paper | Dataset | Twitter | Discord |

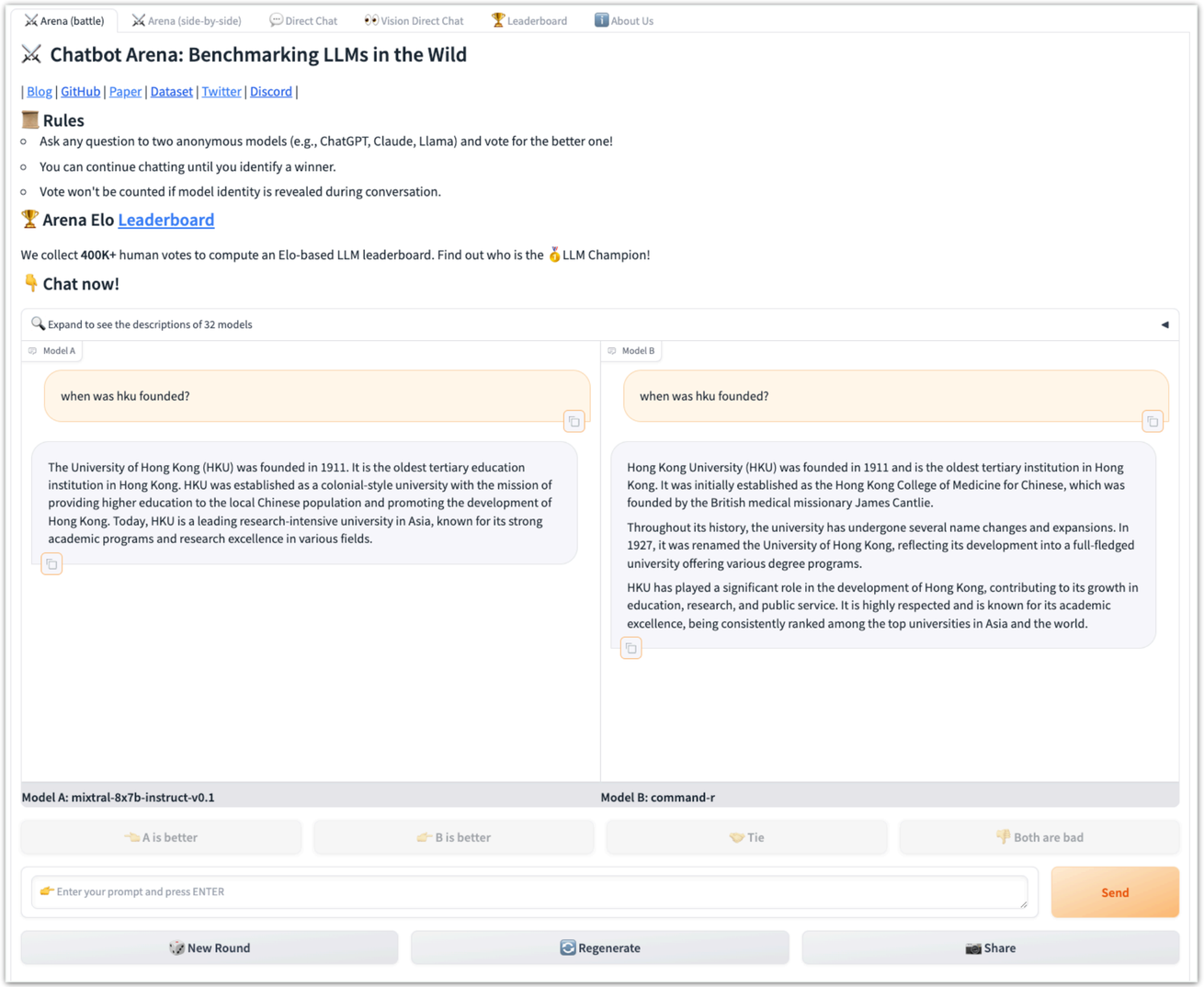
LMSYS Chatbot Arena is a crowdsourced open platform for LLM evals. We've collected over **400,000** human preference votes to rank LLMs with the Elo ranking system.

Arena Elo | Full Leaderboard

Total #models: **73**. Total #votes: **408144**. Last updated: March 13, 2024.

Contribute your vote at chat.lmsys.org! Find more analysis in the notebook.

Rank	Model	Arena Elo	95% CI	Votes	Organization	License	Knowledge Cutoff
1	GPT-4-1106-preview	1251	+5/-4	48226	OpenAI	Proprietary	2023/4
1	GPT-4-0125-preview	1249	+5/-6	22282	OpenAI	Proprietary	2023/12
1	Claude 3 Opus	1247	+6/-6	14854	Anthropic	Proprietary	2023/8
4	Bard (Gemini Pro)	1202	+6/-7	12623	Google	Proprietary	Online
4	Claude 3 Sonnet	1190	+6/-6	14845	Anthropic	Proprietary	2023/8
5	GPT-4-0314	1185	+4/-6	27245	OpenAI	Proprietary	2021/9
7	GPT-4-0613	1159	+4/-5	43783	OpenAI	Proprietary	2021/9
7	Mistral-Large-2402	1155	+5/-6	18959	Mistral	Proprietary	Unknown
8	Qwen1.5-72B-Chat	1146	+4/-5	16729	Alibaba	Qianwen LICENSE	2024/2
8	Claude-1	1145	+5/-6	21929	Anthropic	Proprietary	Unknown
8	Mistral Medium	1145	+5/-4	23931	Mistral	Proprietary	Unknown



Arena (battle) | Arena (side-by-side) | Direct Chat | Vision Direct Chat | Leaderboard | About Us

⚔ Chatbot Arena: Benchmarking LLMs in the Wild

| Blog | GitHub | Paper | Dataset | Twitter | Discord |

📜 Rules

- Ask any question to two anonymous models (e.g., ChatGPT, Claude, Llama) and vote for the better one!
- You can continue chatting until you identify a winner.
- Vote won't be counted if model identity is revealed during conversation.

🏆 Arena Elo Leaderboard

We collect **400K+** human votes to compute an Elo-based LLM leaderboard. Find out who is the 🥇 LLM Champion!

👇 Chat now!

Expand to see the descriptions of 32 models

Model A

when was hku founded?

The University of Hong Kong (HKU) was founded in 1911. It is the oldest tertiary education institution in Hong Kong. HKU was established as a colonial-style university with the mission of providing higher education to the local Chinese population and promoting the development of Hong Kong. Today, HKU is a leading research-intensive university in Asia, known for its strong academic programs and research excellence in various fields.

Model B

when was hku founded?

Hong Kong University (HKU) was founded in 1911 and is the oldest tertiary institution in Hong Kong. It was initially established as the Hong Kong College of Medicine for Chinese, which was founded by the British medical missionary James Cantlie.

Throughout its history, the university has undergone several name changes and expansions. In 1927, it was renamed the University of Hong Kong, reflecting its development into a full-fledged university offering various degree programs.

HKU has played a significant role in the development of Hong Kong, contributing to its growth in education, research, and public service. It is highly respected and is known for its academic excellence, being consistently ranked among the top universities in Asia and the world.

Model A: mixtral-8x7b-instruct-v0.1 | Model B: command-r

👈 A is better | 👉 B is better | 🤝 Tie | 👎 Both are bad

👉 Enter your prompt and press ENTER | Send

🎲 New Round | 🔄 Regenerate | 📷 Share

https://huggingface.co/spaces/lmsys/chatbot-arena-leaderboard

Evaluation: Takeaways

- *Content-overlap metrics* provide a good starting point for evaluating the generation quality, but they're not good enough on their own

- *Model-based metrics* can be more correlated with human judgment, but often are not interpretable

- Human judgments are critical
  - But humans are inconsistent!

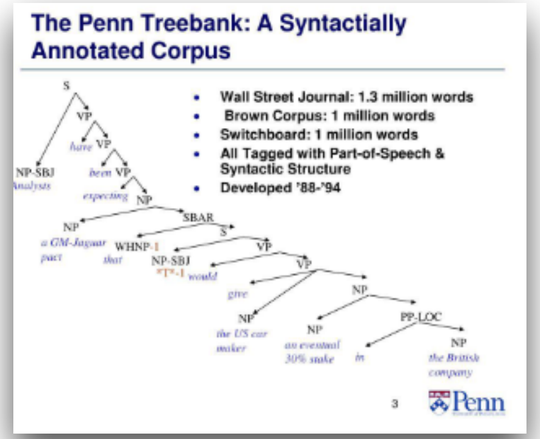
- In many cases, the best judge of output quality is **YOU**!
  - **Look at the actual generations - don't just rely on numbers.**
  - **Publicly release large samples of outputs from your system!**

Everything in evaluation boils down to…

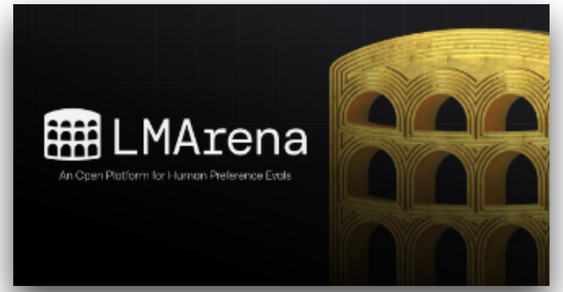
- ***How*** should we evaluate: Evaluation Metrics (what we looked at previously)
- ***What*** should we evaluate?

A primer on Benchmarking

Different eras of benchmarking



SQuAD2.0
The Stanford Question Answering Dataset



IMAGENET

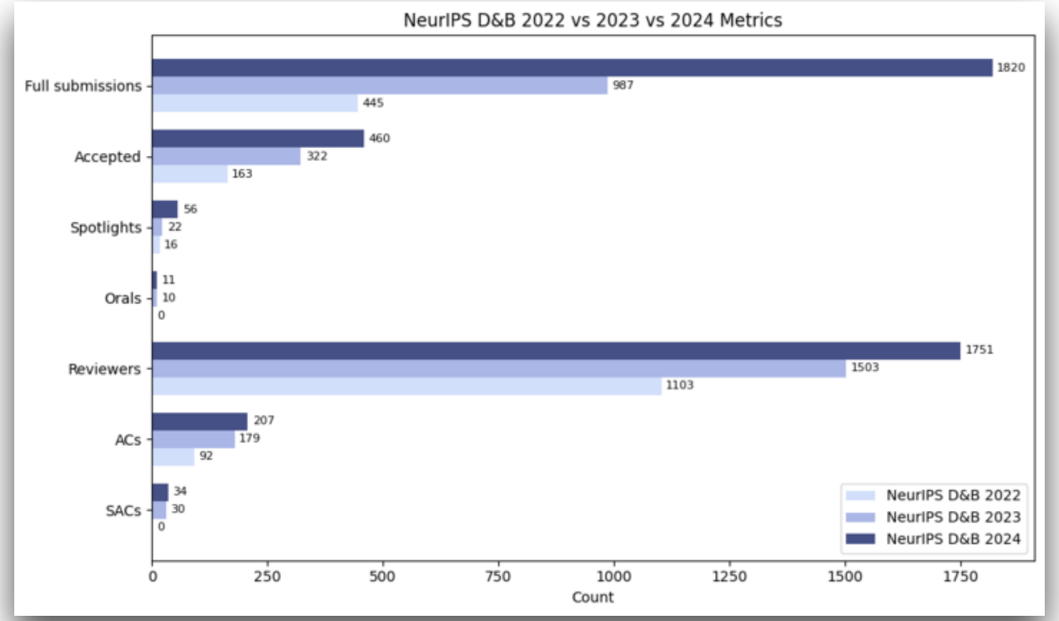
GLUE



SWE-bench

Most common Benchmarks

- Quite a lot in number!
- Keep on increasing…
- Depend on the frontier LLMs vs. Open Source LLMs



	Benchmark		Gemini 4 Argon	GPT-6 Astra	Claude Fable 5.1	Claude Opus 5.5
Knowledge work	Vals Index		68.9%	63.1%	65.8%	67.0%
	AutomationBench	Score	51.3%	41.4%	31.4%	42.5%
	Vals Finance Agent v2		65.4%	53.5%	58.9%	58.6%
	Harvey's Legal Agent Benchmark		19.6%	5.4%	6.7%	3.8%
Agentic coding	DeepSWE v1.1		77.9%	74.1%	67.4%	74.2%
	FrontierSWE v2		55.0%	65.5%	56.3%	62.3%
	Vibe Code Bench		91.9%	89.6%	90.3%	90.3%
	Terminal-bench 4.0		57.4%	58.2%	57.9%	66.4%
ML engineering	PostTrainBench		45.3%	44.3%	40.2%	49.3%
Science and math	Terminal-Bench Science 0.1		57.6%	68.1%	52.6%	63.3%
	LABBench 2		88.8%	85.4%	68.6%	73.1%
	RiemannBench		76.0%	72.0%	65.6%	69.6%
Long context	GraphWalks	Up to 128k, BFS (F1)	99.7%	98.7%	91.4%	90.6%
		256k to 1M, BFS (F1)	84.2%	71.8%	65.0%	66.8%
Computer use	Agent's Last Exam	Pass rate	39.5%	34.2%	—	38.2%
	OSWorld-2.0	Offline subset Partial score	69.2%	72.6%	—	—
Multimodal understanding	Chartography		71.6%	71.0%	46.2%	66.3%
	LVBench		91.7%	87.5%	79.7%	83.7%
Cybersecurity	CWE-bench v1		68.0%	68.0%	58.0%	67.0%

Methodology: deepmind.google/models/evals-methodology/gemini-4-argon

Multilingual QA

- MMMLU: Multilingual extension of Massive Multitask Language Understanding (MMLU)
  - Covering 57 subjects including elementary mathematics, US history, computer science, law, and more, with 15908 questions in total.

Conceptual Physics

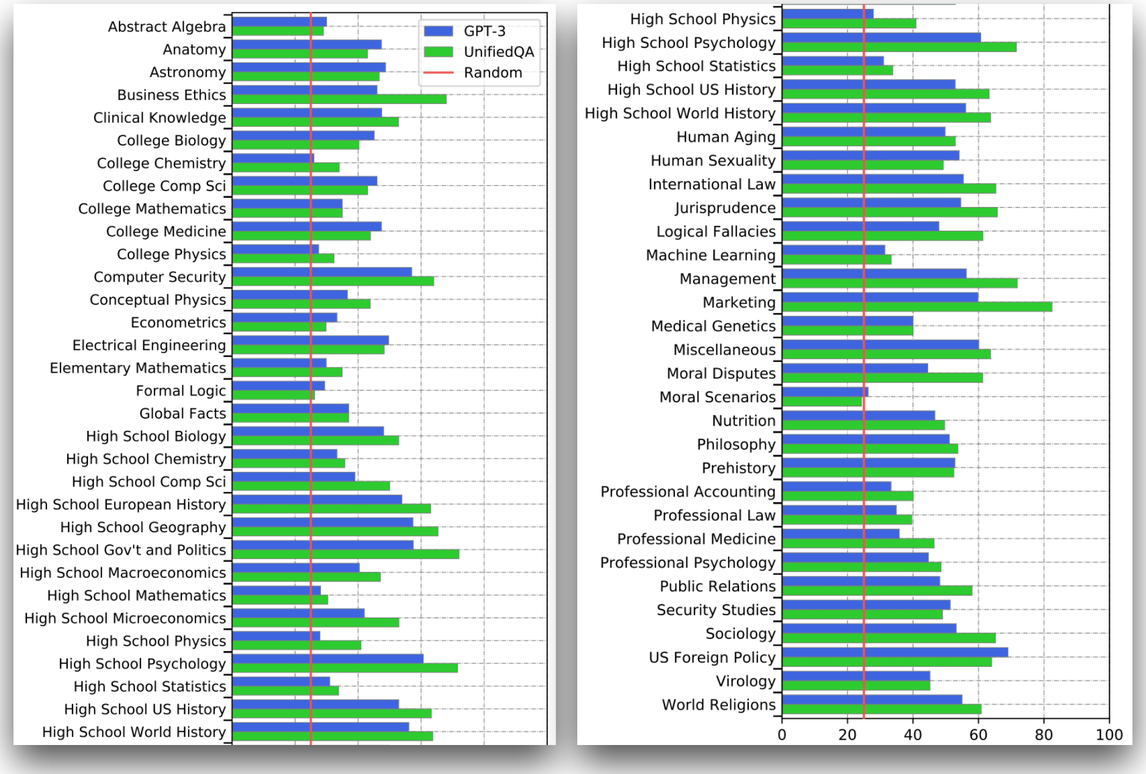
When you drop a ball from rest it accelerates downward at 9.8 m/s². If you instead throw it downward assuming no air resistance its acceleration immediately after leaving your hand is

(A) 9.8 m/s² ✔
(B) more than 9.8 m/s² ✘
(C) less than 9.8 m/s² ✘
(D) Cannot say unless the speed of throw is given. ✘

College Mathematics

In the complex z-plane, the set of points satisfying the equation $z^2 = |z|^2$ is a

(A) pair of points ✘
(B) circle ✘
(C) half-line ✘
(D) line ✔



Visual reasoning

- MMMU: A Massive Multi-discipline Multimodal Understanding and Reasoning Benchmark
  - Including 11.5K meticulously collected multimodal questions from college exams, quizzes, and textbooks, covering six core disciplines.

Art & Design	Business	Science
Question: Among the following harmonic intervals, which one is constructed incorrectly? **Options:** (A) Major third <*image 1*> (B) Diminished fifth <*image 2*> **(C) Minor seventh <*image 3*>** (D) Diminished sixth <*image 4*>	**Question:** ...The graph shown is compiled from data collected by Gallup <*image 1*>. Find the probability that the selected Emotional Health Index Score is between 80.5 and 82? **Options:** (A) 0 (B) 0.2142 **(C) 0.3571** (D) 0.5	**Question:** <*image 1*> The region bounded by the graph as shown above. Choose an integral expression that can be used to find the area of R. **Options:** **(A)** $\int_0^{1.5}[f(x)-g(x)]dx$ (B) $\int_0^{1.5}[g(x)-f(x)]dx$ (C) $\int_0^{2}[f(x)-g(x)]dx$ (D) $\int_0^{2}[g(x)-x(x)]dx$
Subject: Music; **Subfield:** Music; **Image Type:** Sheet Music; **Difficulty:** Medium	**Subject:** Marketing; **Subfield:** Market Research; **Image Type:** Plots and Charts; **Difficulty:** Medium	**Subject:** Math; **Subfield:** Calculus; **Image Type:** Mathematical Notations; **Difficulty:** Easy
Health & Medicine	**Humanities & Social Science**	**Tech & Engineering**
Question: You are shown subtraction <*image 1*>, T2 weighted <*image 2*> and T1 weighted axial <*image 3*> from a screening breast MRI. What is the etiology of the finding in the left breast? **Options:** (A) Susceptibility artifact (B) Hematoma **(C) Fat necrosis** (D) Silicone granuloma	**Question:** In the political cartoon, the United States is seen as fulfilling which of the following roles? <*image* 1> **Option**: (A) Oppressor (B) Imperialist **(C) Savior** (D) Isolationist	**Question:** Find the VCE for the circuit shown in <*image 1*>. Neglect VBE **Answer: 3.75** **Explanation**: …IE = [(VEE) / (RE)] = [(5 V) / (4 k-ohm)] = 1.25 mA; VCE = VCC - IERL = 10 V - (1.25 mA) 5 k-ohm; VCE = 10 V - 6.25 V = 3.75 V
Subject: Clinical Medicine; **Subfield:** Clinical Radiology; **Image Type:** Body Scans: MRI, CT.; **Difficulty:** Hard	**Subject:** History; **Subfield:** Modern History; **Image Type:** Comics and Cartoons; **Difficulty:** Easy	**Subject:** Electronics; **Subfield:** Analog electronics; **Image Type:** Diagrams; **Difficulty:** Hard

Math problem-solving

- MATH 500: 500 math problems- spanning topics like probability, algebra, trigonometry, and geometry.
- AIME 2024: Problems from the American Invitational Mathematics Examination (a prestigious high school mathematics competition) 2024

Problem string · *lengths* 333↔405 23.3%	Solution string · *lengths* 284↔657 43.3%	Answer int64 23↔110 33.3%
Let x,y and z be positive real numbers that satisfy the following system of equations: \[\log_2\left({x \over yz}\right) = {1 \over 2}\] \[\log_2\left({y \over xz}\right) = {1 \over 3}\] \[\log_2\left({z \over xy}\right) = {1 \over 4}\] Then the value of $\left\|\log_2(x^4y^3z^2)\right\|$ is $\tfrac{m}{n}$ where m and n are relatively prime positive integers. Find $m+n$.	Denote $\log_2(x) = a$, $\log_2(y) = b$, and $\log_2(z) = c$. Then, we have: $a-b-c = \frac{1}{2}$, $-a+b-c = \frac{1}{3}$, $-a-b+c = \frac{1}{4}$. Now, we can solve to get $a = \frac{-7}{24}$, $b = \frac{-9}{24}$, $c = \frac{-5}{12}$. Plugging these values in, we obtain $\|4a + 3b + 2c\| = \frac{25}{8} \implies \boxed{033}$.	33

Convert the point $(0,3)$ in rectangular coordinates to polar coordinates. Enter your answer in the form $(r,\theta),$ where $r > 0$ and $0 \le \theta < 2 \pi.$	We have that $r = \sqrt{0^2 + 3^2} = 3.$ Also, if we draw the line connecting the origin and $(0,3),$ this line makes an angle of $\frac{\pi}{2}$ with the positive x-axis.
	
	[asy]
	unitsize(0.8 cm);
	
	draw((-0.5,0)--(3.5,0));
	draw((0,-0.5)--(0,3.5));
	draw(arc((0,0),3,0,90),red,Arrow(6));
	
	dot((0,3), red);
	label("$(0,3)$", (0,3), W);
	dot((3,0), red);
	[/asy]
	
	Therefore, the polar coordinates are $\boxed{\left(3, \frac{\pi}{2} \right)}.$
		\left(3, \frac{\pi}{2} \right)

Measuring Mathematical Problem Solving With the MATH Dataset

https://huggingface.co/datasets/HuggingFaceH4/MATH-500

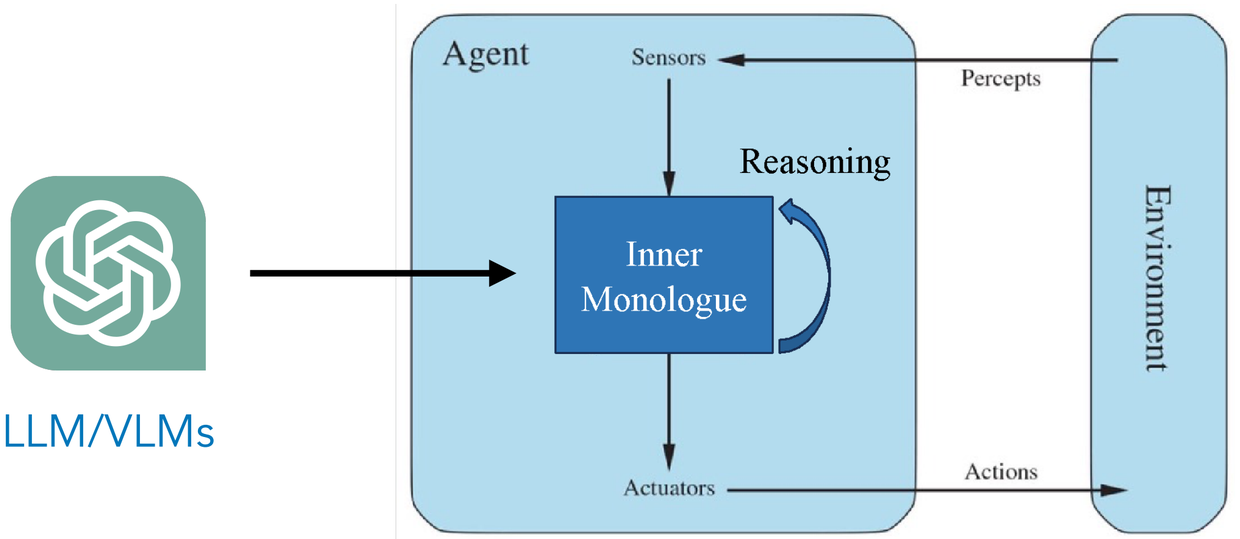
Graduate-level reasoning

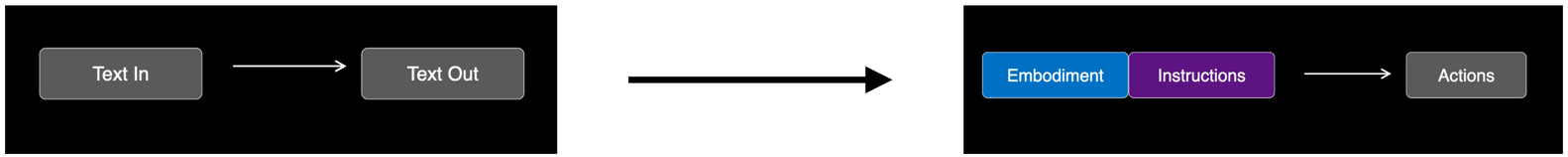
- GPQA
  - a challenging dataset of 448 multiple-choice questions written by domain experts in biology, physics, and chemistry.

Chemistry (general)
A reaction of a liquid organic compound, which molecules consist of carbon and hydrogen atoms, is performed at 80 centigrade and 20 bar for 24 hours. In the proton nuclear magnetic resonance spectrum, the signals with the highest chemical shift of the reactant are replaced by a signal of the product that is observed about three to four units downfield. Compounds from which position in the periodic system of the elements, which are also used in the corresponding large-scale industrial process, have been mostly likely initially added in small amounts? A) A metal compound from the fifth period. B) A metal compound from the fifth period and a non-metal compound from the third period. C) A metal compound from the fourth period. D) A metal compound from the fourth period and a non-metal compound from the second period.
Organic Chemistry
Methylcyclopentadiene (which exists as a fluxional mixture of isomers) was allowed to react with methyl isoamyl ketone and a catalytic amount of pyrrolidine. A bright yellow, cross-conjugated polyalkenyl hydrocarbon product formed (as a mixture of isomers), with water as a side product. These products are derivatives of fulvene. This product was then allowed to react with ethyl acrylate in a 1:1 ratio. Upon completion of the reaction, the bright yellow color had disappeared. How many chemically distinct isomers make up the final product (not counting stereoisomers)? A) 2 B) 16 C) 8 D) 4

GPQA: A Graduate-Level Google-Proof Q&A Benchmark

LLM/VLMs as agents

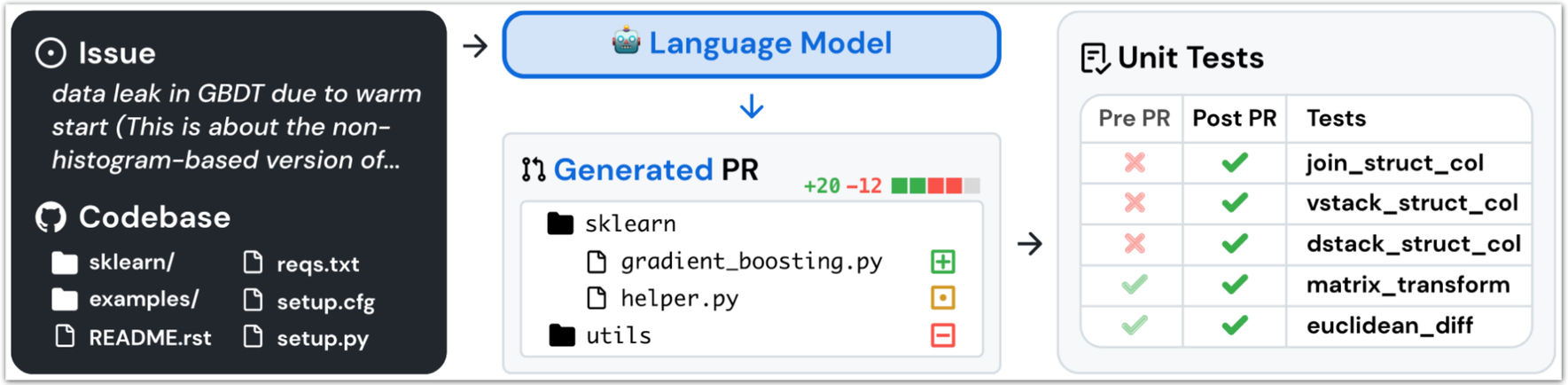




https://openai.com/index/computer-using-agent/

Agents coding

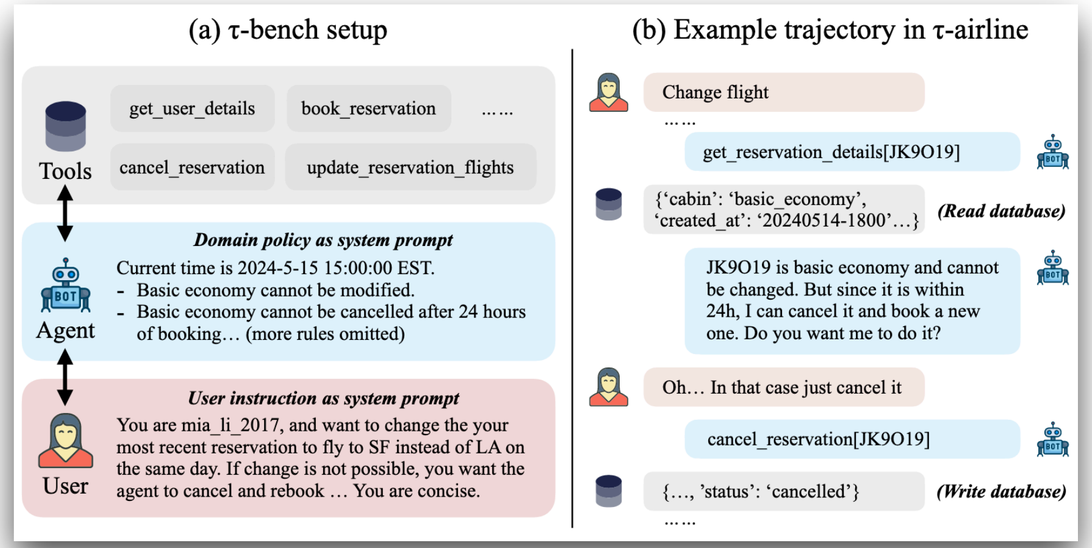
- SWE-bench: Project-level coding tasks
  - Environment: project code repos, filesystems, IDEs…
  - Observation space: code files, running outputs, docs, errors/issues, commit history…
  - Action space: code edits, file search/view, test updates…



SWE-bench: Can Language Models Resolve Real-World GitHub Issues?, (Jimenez et al., 2023)
Spider 2.0: Evaluating Language Models on Real-World Enterprise Text-to-SQL Workflows, (Lei et al., 2024) MLE-bench: Evaluating Machine Learning Agents on Machine Learning Engineering, (Chan et al., 2024)

Agentic tool use

- TAU-bench: API/functional calling for tool use
  - Environment: software systems such as databases, app/web services…
  - Observation space: API docs, system info, error messages and logs…
  - Action space: function calls, error handling routines…





```
{"order_id": "#W2890441",
"user_id": "mei_davis_8935",
"items": [{
    "name": "Water Bottle",
    "product_id": "8310926033",
    "item_id": "2366567022",
    "price": 54.04,
    "options": {
        "capacity": "1000ml",
        "material": "stainless
        steel",
        "color": "blue"
    }}, ...], ...}
```

(a) An `orders` database entry in τ-retail.

```
def return_delivered_order_items(
    order_id: str,
    item_ids: List[str],
    payment_method_id: str,
) -> str: ...

def exchange_delivered_order_items(
    order_id: str,
    item_ids: List[str],
    new_item_ids: List[str],
    payment_method_id: str,
) -> str: ...
```

(b) An API tool in τ-retail.

```
## Return delivered order
- After user confirmation, the order status
will be changed to 'return requested'...

## Exchange delivered order
- An order can only be exchanged if its
status is 'delivered'...
```

(c) Domain policy excerpts in τ-retail.

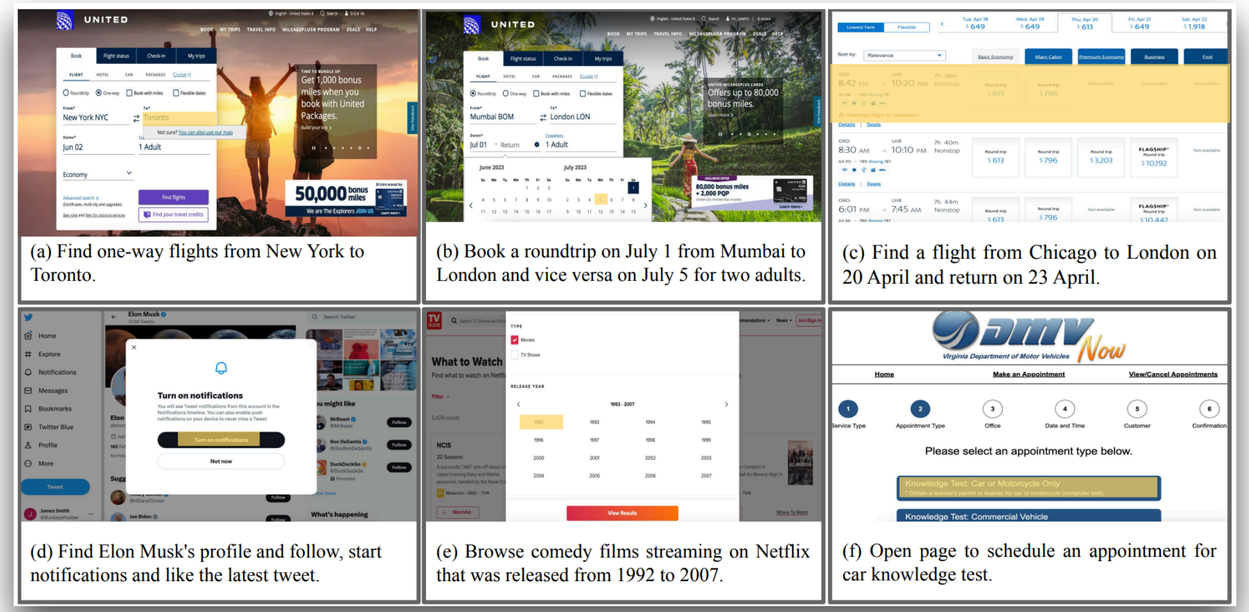
```
{"instruction": "You are Mei Davis in 80217.
You want to return the water bottle, and
exchange the pet bed and office chair to the
cheapest version. Mention the two things
together. If you can only do one of the two
things, you prefer to do whatever saves you
most money, but you want to know the money
you can save in both ways. You are in debt
and sad today, but very brief.",
"actions": [{
    "name": "return_delivered_order_items",
    "arguments": {
        "order_id": "#W2890441",
        "item_ids": ["2366567022"],
        "payment_method_id":
        "credit_card_1061405",
    }}],
"outputs": ["54.04", "41.64"]}
```

(d) User instruction ensures only one possible outcome.

τ-bench: A Benchmark for Tool-Agent-User Interaction in Real-World Domains, (Yao et al., 2024)

Web/app agents

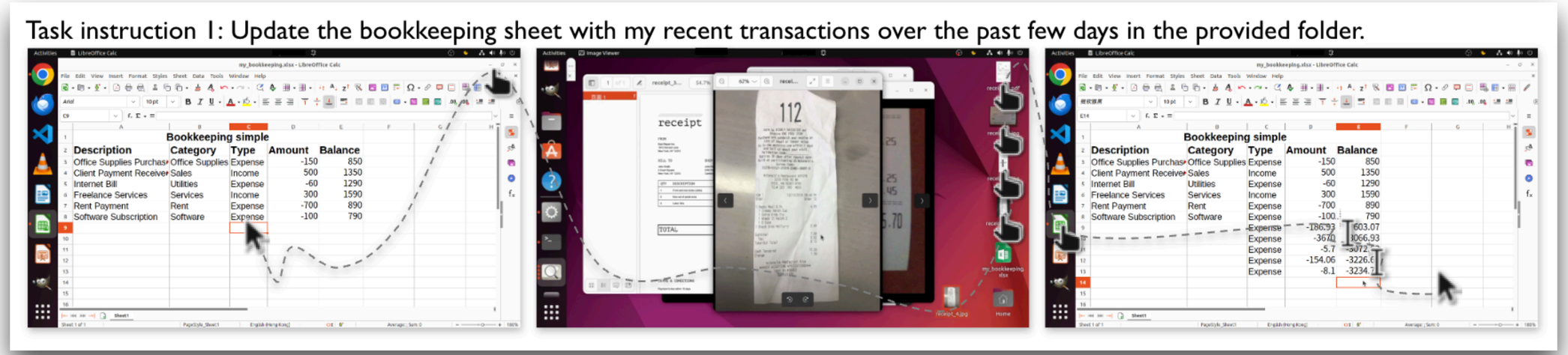
- WebArena: LLM/VLMs as agents for web/app use
  - Environment: web browsers/apps
  - Observation space: screenshots, DOM trees, HTML, historical actions…
  - Action space: browser/app controls (e.g., click, type, scroll, drag, hover…)



World of Bits: An Open-Domain Platform for Web-Based Agents, (Shi et al., 2017)
Mind2Web: Towards a Generalist Agent for the Web, (Deng et al., 2023)
WebArena: A Realistic Web Environment for Building Autonomous Agents, (Zhou et al., 2023)

Computer use agents

- OSWorld: Computer use for universal digital tasks
  - Environment: desktop operating systems
  - Observation space: desktop screenshots, a11y trees, historical actions…
  - Action space: keyboard/mouse controls (e.g., click, type, drag, shortcuts)



Try it out: https://arena.xlang.ai

OpenAI Universe, (OpenAI, 2016)
OSWorld: Benchmarking Multimodal Agents for Open-Ended Tasks in Real Computer Environments, (Xie et al., 2024)

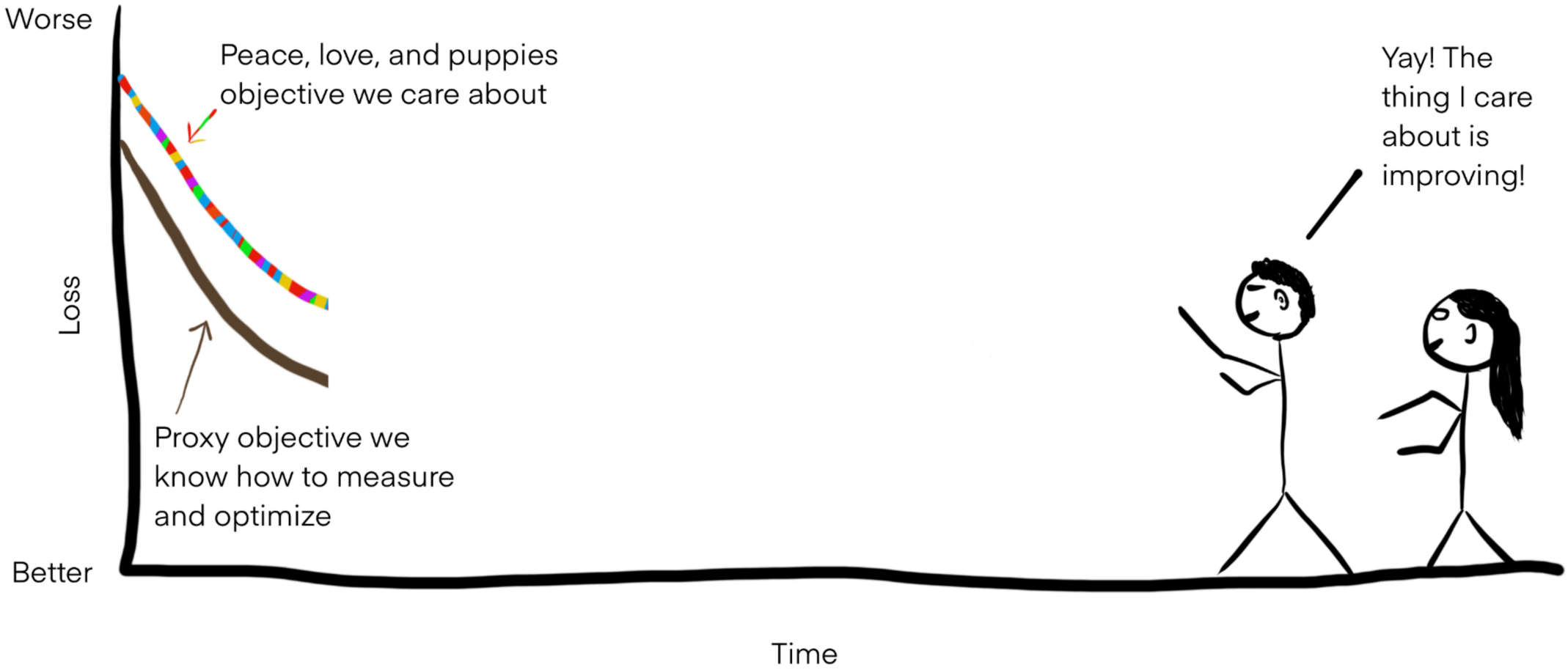
Everything in evaluation boils down to…

- ***How*** should we evaluate: Evaluation Metrics
- ***What*** should we evaluate: Benchmarks and Tasks

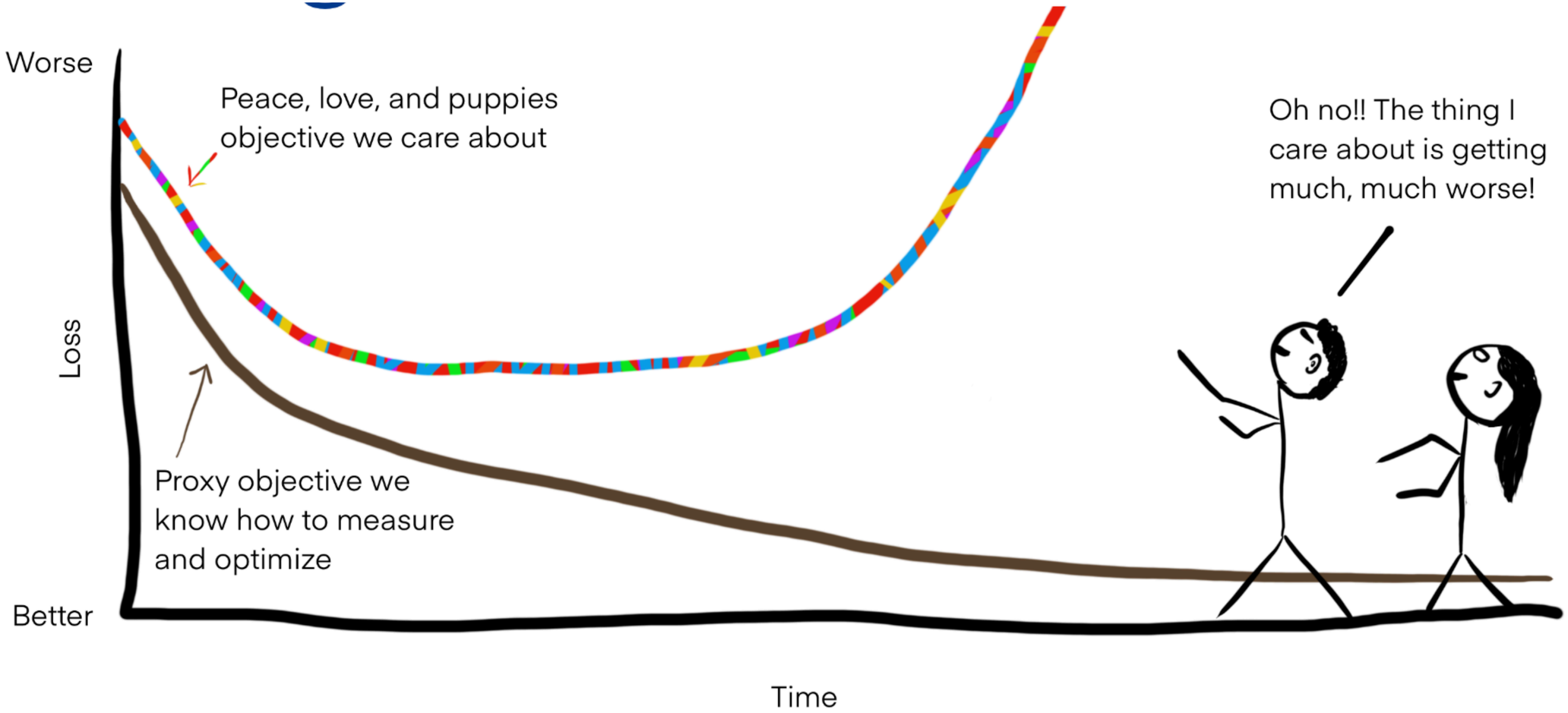
Goodhart's laws: "when a measure becomes a target, it ceases to be a good measure"



Goodhart's laws: "when a measure becomes a target, it ceases to be a good measure"



Goodhart's laws: "when a measure becomes a target, it ceases to be a good measure"



Concluding Thoughts

- Interacting with NLG systems quickly shows their limitations
- Even in tasks with more progress, there are still many improvements ahead
- Evaluation remains a huge challenge
  - We need betters ways to automatically evaluate NLG systems
- One of the most exciting areas of NLP to work in!